Resilience of Global Supply Chains and Generic Drug Shortages

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Since 2009, U.S. hospitals have faced persistent shortages of generic injectable drugs, while their production has shifted toward the Global South. To assess the role of offshoring, I construct a new dataset mapping these drugs to their manufacturing plants and show that offshoring increases shortage incidence and duration. To explain this relationship, I develop and estimate a model of global procurement that quantifies drug manufacturers' trade-off between cost-cutting and supply reliability. Limited transparency makes reliability difficult to reward, while prices and capacity adjust slowly after disruptions. Offshoring magnifies these pre-existing frictions by expanding access to cheaper but riskier plants. To quantify unmet demand and its welfare cost, I develop a censored-demand estimator and derive realized consumer surplus under rationing in closed form. Counterfactuals show that reshoring subsidies and price premia during shortages do little to reduce unmet demand. A failure-to-supply penalty of half the contract price per missing unit induces larger buffers, reducing the expected share of unmet demand by 14 percent while raising contract prices by 9 percent. Improved access nearly offsets the losses from higher prices, with modest gains in consumer surplus when fines are rebated. Procurement contracts that reward resilient capacity before disruptions can therefore reduce shortages without reversing offshoring.

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"We've paid a high price for the low costs we enjoy (...) The White House is talking about more regulation, generic manufacturers are calling for direct subsidies. (...) We need to create a market for high quality manufacturing, so you can allow generic manufacturers to make certain claims about the reliability of the manufacturing. And then (...) you pay them for the fact that they have reliable manufacturing that might be domestic, that might be more modern, so it's going to be less prone to shortages."

Scott Gottlieb, former FDA commissioner, on May 21, 2023.

1. Introduction

One should not expect shortages to persist in competitive markets, yet they have become commonplace in the generic injectable pharmaceutical market.[1] Pharmaceutical shortages are conditions of excess demand: at prevailing prices, available supply is insufficient to meet patients' demand. Early signs of vulnerability appeared around 2009 in the United States, when shortages of injectable oncology drugs began to surge; in 2011 alone, 267 drugs were newly reported in shortage. Even more striking than this initial surge is the *persistence* of shortages: drugs can remain in short supply for months or even years (Figure 1). Nor is the problem confined to the United States (Additional Results Section S3.2). In 2018, the American Medical Association declared drug shortages an urgent public health crisis and urged the federal government to treat them as a national security concern.[2]

Short-run rigidities help explain why a disruption in production can leave demand unmet for so long. Procurement contracts limit price adjustments, while specialized sterile production capacity is costly and slow to expand. Yet these frictions have long been features of injectable drug markets. Notably, the surge in shortages coincided with a broader shift of pharmaceutical manufacturing abroad, increasingly to the Global South. Among plants registered to supply the U.S. pharmaceutical market, the share located abroad rose from about 15 percent in the early 2000s to more than 80 percent by 2020.[3] Concerns about dependence on foreign suppliers have prompted calls to bring drug production back to the United States (Appendix C.2). However, aggregate manufacturing statistics do not establish whether the drugs in shortage are disproportionately produced abroad. To what extent is the offshoring of generic drug manufacturing plants to the Global South contributing to U.S. shortages?

I argue that offshoring magnifies pre-existing frictions in pharmaceutical procurement. Access to cheaper but riskier plants widens differences in supply reliability, yet procurement contracts have not adapted to reward more reliable suppliers.[4] I address this question in three steps. First, I construct a novel dataset linking U.S. injectable drug products to their manufacturing plants and provide evidence

[1] Injectable drugs are administered by injection or infusion rather than taken orally. They include cancer treatments, antibiotics, and anesthetics. Most injectable drugs are off-patent products operating in contestable markets (see Additional Results Section S1.4 for details).

[2] See the AMA's June 12, 2018 statement, "Drug shortages a matter of national security", and the recommendations from the 2018 summit convened by ASHP and other health organizations, "Drug Shortages as a Matter of National Security". The problem persisted beyond the study period: ASHP's August 2023 report recorded 309 ongoing shortages, the highest number in nearly a decade.

[3] Author's calculations from establishment-registration records of the FDA's Center for Drug Evaluation and Research (CDER). These figures cover drug-manufacturing plants across all dosage forms, not only injectables.

[4] Buyers (hospitals and their group purchasing organizations) choose among therapeutically interchangeable generics with

that offshoring increases the frequency and persistence of shortages. Second, I develop and estimate a partial-equilibrium model of global drug procurement in which manufacturers trade off production costs and supply reliability. The model generates shortages in equilibrium through manufacturers' location and capacity choices. I develop an estimation method that recovers demand when shortages constrain sales, allowing me to quantify unmet demand and its welfare cost. Third, I use the estimated model to compare reshoring subsidies with changes in procurement contracts at existing production locations. Penalties for missing deliveries induce larger capacity buffers and reduce shortages more effectively than reshoring subsidies or allowing prices to adjust to supply conditions. These availability gains come at a cost: the value of fewer shortages must be weighed against the higher contract prices needed to support reliable capacity.

The first part of the paper maps the global supply chains of injectable drugs sold in the United States. Doing so requires overcoming a longstanding traceability problem: existing sales data identify drug labelers, the companies marketing the drugs, but not the plants that manufacture them. Without these links, it is difficult to establish where shortages originate or to assess plant-level supply risk.[5] I recover these links by combining FDA records of domestic and foreign manufacturing facilities with detailed U.S. and foreign customs data. I extract drug names from shipment descriptions and match shippers to registered plants by name and address. I also digitize thousands of supplier-level availability reports and link them to drug characteristics, prices, and national sales. The resulting panel links injectable products, both on and off the shortage lists, to their labelers, plants, and shortage histories from the early 2000s to 2019.[6] Throughout the paper, offshoring refers to production location, regardless of whether the plant is owned by the labeler or operated by a contract manufacturing organization.

Using these data, I document that drugs experiencing shortages are more heavily offshored, and that shortage onset coincides with a sharp decline in the number of foreign suppliers shipping the affected drug to the United States. I also find that drugs whose predicted demand grows in Asia are more likely to be manufactured there and to experience shortages in the United States. The production response reflects a home-market effect: growing demand for particular treatments attracts their manufacturing to these markets (Krugman, 1980). I exploit this demand-driven variation to address selection into offshoring, using the disease burden predicted by these countries' demographic composition and insurance coverage as a shift-share instrument (Acemoglu and Linn, 2004; Costinot et al., 2019).[7]

To strengthen the evidence for a causal interpretation, I use a dynamic difference-in-differences design that traces how shortages evolve in the years following offshoring. I compare changes in shortages

limited information about plant-level reliability. Suppliers (drug labelers, the firms marketing the products) lose revenue on units they fail to deliver, but this private loss need not reflect the full cost of unmet demand to buyers. Section 2 describes these procurement and information frictions (FDA, 2019a; FDA, 2019b).

[5]Hospitals and wholesalers cannot always identify which plants make their drugs or the outsourcing relationships involved, since these arrangements are treated as commercially confidential (Department of Homeland Security, 2018). The FDA's Drug Shortages Task Force notes that the pharmaceutical industry "regards the location of their manufacturing facilities as confidential commercial information" (FDA, 2019a, p. 44). The FDA also reports that its own data do not measure how much of a drug is produced at each facility (FDA, 2019a, p. 117).

[6]I link 95 percent of injectable products on the shortage list to at least one manufacturing plant (Appendix F.2).

[7]The IV estimate implies that a shift from entirely domestic to entirely Asian registered manufacturing plants raises the annual probability of a new shortage by about 30 percentage points (Appendix Table E.6).

before and after a drug is offshored with changes over the same years among drugs not yet offshored. I repeat the comparison using only drugs that remain domestic throughout (Callaway and Sant'Anna, 2021). Averaged over the offshoring year and the following decade, the probability of being in shortage rises by about 9 percentage points in the first comparison and 13 percentage points in the second, against an average shortage rate of 27 percent among offshored drugs. The corresponding point estimates for the shortage-length measure are about 70 and 120 days (Appendix Table E.5). In the baseline comparison, the increase in shortage risk becomes pronounced from the fourth year after offshoring and persists through the tenth (Figure 4). This delayed and persistent response is consistent with a lasting deterioration in supply reliability, rather than merely a temporary interruption during relocation.

What, then, motivates manufacturers to let supply periodically fail? The reduced-form evidence does not identify the costs of more reliable supply or how firms would respond to different procurement rules. In the second part of the paper, I develop a structural model of manufacturers' location and production choices in global procurement markets. Manufacturers choose a production location and then set bid prices and capacity before supply shocks occur. U.S. hospital buyers allocate purchases according to suppliers' prices and product characteristics. Prices, capacity, and purchase allocations remain fixed within the contract period, so a shortage occurs when realized supply falls below contracted demand.

Manufacturers have two available margins to insure against shortage risks: they can build capacity buffers above contracted quantities and choose production locations with lower disruption risk. Offshoring gives firms access to lower-cost but riskier plants, while larger buffers reduce shortage risk at an additional production cost. The trade-off between supply reliability and cost-cutting thus appears in both stages of the firm's problem. Although shocks are exogenous conditional on location, shortage risk depends on manufacturers' choices. Suppliers weigh these costs against expected revenue losses from unfulfilled contracts, but do not bear the full cost of unmet demand to buyers. The model quantifies this trade-off and generates shortages in equilibrium. It also provides a framework to evaluate alternative contract designs and location policies in terms of prices, drug availability, and welfare.

Shortages introduce a censoring problem for demand estimation: for a supplier in shortage, observed sales reflect realized supply and provide only a lower bound on demand. A standard demand inversion would attribute missing sales to low demand and distort the implied substitution patterns. To address this problem, I develop a two-stage procedure that uses sales from both shortage and non-shortage periods. I first estimate the demand coefficients using a within-market transformation of the logit model, without specifying potential market size. I then jointly recover market size and the distribution of demand shocks through an iterative censored-likelihood procedure, and use them to estimate latent quantities and market shares.[8] The resulting estimates quantify the unmet demand that shortage records alone cannot measure.

[8]The within-market transformation follows Huang and Rojas (2014) and Dubois and Lasio (2018). The demand coefficients are initially estimated from product-years without reported shortages. Because shortage status depends on demand, subsequent iterations also include product-years with shortages, replacing their observed sales with imputed demand. Within each iteration, market size is calibrated so that model-implied demand for products without shortages matches their aggregate observed sales; the censored likelihood estimates the distribution of demand shocks. Section 5.2 and Appendix B.4 describe the procedure and its assumptions.

The distinction between demand and realized sales also matters for welfare. Buyers choose a drug ex ante on the basis of its price and characteristics. After a disruption, some cannot obtain their chosen drug and are forced into an outside option they had rejected. I derive a closed-form expression for realized consumer surplus that accounts for the utility loss conditional on that initial choice. It equals the standard logit surplus minus the share of buyers displaced by shortages, valued at their average utility loss. Changes in realized surplus then decompose into an access effect and a price-and-characteristics effect. A policy can therefore raise consumer surplus even if prices rise, provided the gains from improved access are large enough.[9]

The demand estimates imply high price sensitivity in procurement choices, while separate reduced-form evidence suggests limited market discipline after shortages.[10] On the supply side, I combine recovered demand with shortage outcomes and manufacturers' optimality conditions to estimate the geographical distribution of supply risk and production costs. A location-choice model then identifies fixed-cost differences across production regions from observed plant locations. Asian plants produce at lower cost but face larger losses of effective capacity. U.S. plants face more favorable yield distributions, but capacity is more expensive and buffers are thinner relative to contracted demand. These estimates explain why moving production to a safer location need not deliver a corresponding improvement in equilibrium availability.

Which policy instruments could shift the market toward more reliable supply, and at what cost? I compare two approaches: subsidizing a return to domestic production and changing procurement contracts at existing production locations. The contracting reforms either allow prices to rise during shortages, through a premium on delivered units, or impose penalties on missing deliveries.[11]

Reshoring subsidies substantially reallocate production to the United States but do little to reduce shortages. Motivated by the 2020 U.S. proposal for a domestic drug-manufacturing tax credit, I simulate a subsidy of 10.5 percent of domestic product revenue. At this rate, the average price of products that relocate rises by about 26 percent. Supplier shortage probability is nearly unchanged, while the expected share of unmet demand rises by about 5 percent. The subsidy changes where drugs are made, not how procurement rewards reliable delivery. Consumer surplus nevertheless increases modestly, driven by price reductions in a few large markets rather than improved access.[12]

Allowing prices to rise during shortages creates mixed incentives for capacity investment. The rule pays a premium on every unit delivered whenever at least one supplier in the market is short. A larger buffer allows a supplier to deliver more during a rival's shortage, but can also prevent its own shortfall and eliminate the premium when no other supplier is short. Most suppliers respond by increasing their buffers, but some choose to hold no spare capacity. At a 25 percent premium, supplier

[9] Surplus is measured in expected-utility units. The price-and-characteristics effect also includes changes in the value of avoiding rationing. Sections 4.6 and 6.2 derive the expression and decomposition; Appendix C.1.4 describes policy transfers and price-equivalent welfare changes.

[10] Most suppliers remain active after a shortage, and I find no evidence of a subsequent decline in national market shares among products whose supply has recovered (Appendix D.4).

[11] Manufacturers anticipate each policy when choosing prices and capacity. The simulations use the same estimated parameters and a common 2017–2019 baseline. Only the reshoring counterfactual allows production locations to change.

[12] At this rate, realized consumer surplus rises by 2.8 percent when buyers finance the subsidy (Section 6.3).

shortage probability rises by about a third, while the expected share of unmet demand is essentially unchanged. Suppliers offer lower contract prices in anticipation of premium revenue, but the resulting gain in consumer surplus at contract prices is offset by buyers' premium payments. Paying more for deliveries during shortages is therefore not equivalent to rewarding the capacity that prevents them.

Failure-to-supply penalties instead raise the private cost of missing deliveries and induce larger capacity buffers before shocks occur. A penalty equal to half the contract price per missing unit reduces supplier shortage probability by about 28 percent and the expected share of unmet demand by 14 percent. The additional capacity comes at higher contract prices, which rise by 9 percent when weighted by demand. The gains from improved access nearly offset the losses from higher contract prices, and consumer surplus rises modestly once fines are returned to buyers. At the rates compared here, the penalty is the only policy that substantially reduces both shortage incidence and unmet demand. Taken together, the counterfactuals favor procurement contracts that strengthen ex-ante incentives for resilient capacity over reshoring alone or reliance on price increases after disruptions. Evidence that offshoring contributes to shortages need not imply that reversing it is the most effective remedy. Adapting procurement to reward reliable delivery can improve availability while preserving the production advantages of global supply chains.

Related Literature. This paper contributes to four related strands of research on global production, supply-chain resilience, demand and welfare under supply constraints, and pharmaceutical markets.

First, the paper contributes to the trade literature on the organization of global value chains (Antràs and Chor, 2013; Alfaro et al., 2019; Costinot, Vogel, and Wang, 2013) and offshoring (Grossman and Rossi-Hansberg, 2008; Antràs and Staiger, 2012; Baldwin and Venables, 2013). It is particularly related to research on foreign sourcing and the separation of manufacturing from firms' other activities (Bernard et al., 2020; Antràs et al., 2026; Fort, 2023). In pharmaceutical markets, the firm marketing a drug need not manufacture it, so identifying that firm's nationality does not establish where production takes place. I construct product–plant links that make production geography observable and use them to estimate how offshoring affects shortages.[13] The paper thus examines how the cost advantages of global sourcing interact with procurement contracts that do not fully reward supply reliability.

Second, the paper contributes to research on supply-chain resilience. Models of network formation, diversification, and resilience investment analyze how firms protect production against disruption (Elliott, Golub, and Leduc, 2022; Grossman, Helpman, and Lhuillier, 2023; Grossman, Helpman, and Sabal, 2024). Related empirical work documents the transmission of shocks through production networks and the resilience of supplier relationships (Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho et al., 2021; Barrot and Sauvagnat, 2016; Khanna, Morales, and Pandalai-Nayar, 2022), while Castro-Vincenzi (2022) studies plant location and capacity responses to climate risk. I contribute a structural estimation framework linking product-level prices, sales, manufacturing locations, and shortage histories to manufacturers' location and capacity decisions. By recovering demand when sales are supply-constrained and estimating the geographical distribution of production risk and costs, I quantify the trade-off between

[13] This mapping also contributes to efforts to document global supply networks (Pichler et al., 2023). The focus on off-patent products connects the setting to work on patent protection and the location of production (Bilir, 2014).

cost-cutting and supply reliability under existing procurement contracts. The estimated model then allows me to compare reshoring with procurement reforms that change the private returns to reliable delivery.[14]

Third, the paper contributes to the industrial organization literature on demand estimation and welfare when supply constraints bind. Related work estimates demand under incomplete product availability (Conlon and Mortimer, 2013) and excess demand in capacity-constrained nursing homes (Ching, Hayashi, and Wang, 2015). I study supply-driven shortages: production shocks reduce manufacturers' ability to fulfill demand after prices and capacity have been chosen. Because buyers cannot reoptimize their purchases within the contracting period, these shocks create partial rationing rather than a new choice among available products. I combine reported shortage status with sales data in a censored-demand procedure, building on the treatment of unobserved market size in Huang and Rojas (2014) and Dubois and Lasio (2018). I also derive a closed-form adjustment to the standard logit surplus (Anderson, de Palma, and Thisse, 1992; Capps, Dranove, and Satterthwaite, 2003), valuing the losses of rationed buyers conditional on their initial choices. Together, these methods recover unmet demand and separate the welfare effects of improved availability from those of prices and product characteristics.

Finally, the pharmaceutical application builds on studies of generic entry and price competition (Scott Morton, 1999; Scott Morton, 2000; Frank and Salkever, 1997; Berndt, Conti, and Murphy, 2017; Conti and Berndt, 2020; Frank, Hicks, and Berndt, 2021) and of the organization and geography of generic production (Berndt, Conti, and Murphy, 2018; Kaygisiz et al., 2019). The analysis also relates to research on the economic causes and consequences of drug shortages (Woodcock and Wosińska, 2013; Stomberg, 2018; Yurukoglu, Liebman, and Ridley, 2017; Conti and Wosińska, 2025; Dubois, Majewska, and Reig, 2026). I add evidence linking shortages to specific manufacturing locations and estimate how location, price, and capacity choices respond to procurement policies. The paper also relates to Atal, Cuesta, and Sæthre (2026), whose analysis of quality certification shows how information and compliance costs jointly affect demand, entry, and prices in generic pharmaceuticals.[15] Here, the relevant dimension of supplier performance is reliable delivery: therapeutically interchangeable drugs need not have equally reliable supply, and procurement affects the prices and availability buyers ultimately face.

Outline. The paper proceeds as follows. Section 2 describes the market and the data used to map injectable drugs supply chains. Section 3 provides new empirical evidence on the links between offshored production and shortage spells. Section 4 builds a model in which shortages arise endogenously in global drug markets. Section 5 estimates the model. Section 6 evaluates counterfactual policies. The last section concludes. The Appendix and the Additional Results provide supporting evidence, methodological details, and proofs.

[14]The emphasis on procurement incentives also relates to Taylor and Wiggins (1997), who compare supplier incentives under competitive bidding and long-term subcontracting relationships, and Heise et al. (2024), who study how trade policy uncertainty shapes the use of such relationships in international sourcing.

[15]Related work examines patent protection, parallel trade, and international pricing (Chaudhuri, Goldberg, and Jia, 2006; Dubois and Sæthre, 2020; Grossman and Lai, 2008; Dubois, Gandhi, and Vasserman, 2022; Ganapati and McKibbin, 2019).

2. Market Structure: Tracing Injectable Drugs' Supply Chains

This section describes the drugs affected by shortages, the organization of their procurement and production, and the data I construct to map their supply chains.

2.1. Generic Injectable Drugs and Shortages

The Case of Generic Sterile Injectable Drugs. The drug shortage problem almost entirely affects generic drugs, which represent 90 percent of prescriptions in the U.S., and in particular sterile injectables. Sterile injectables include many cancer treatments, antibiotics, and anesthetics, and account for 80 percent of U.S. drug shortages, despite representing only 29 percent of the generics market in volume. I focus through this paper on this generic sterile injectables segment. These are predominantly older, off-patent medicines (Additional Results Section S2.1), administered by injection or infusion to patients in hospitals. Many have few clinical substitutes, and treatment cannot easily be postponed. Within this segment, more than half of the generic injectable drugs in the Orange Book data were reported in shortage at least once between 2004 and 2019 (Appendix Figure D.2a).

Defining Shortages: Theory versus Empirics. The Food and Drug Administration (FDA) defines a drug shortage as insufficient aggregate supply of clinically interchangeable versions of a drug to meet current or projected patient demand (Haninger, Jessup, and Koehler, 2011; FDA, 2019a). A shortage is thus a condition of *excess demand*, rather than simply a decline in sales or an interruption at one plant. My shortage records were provided by the University of Utah Drug Information Service (UUDIS).[16] These records identify the affected drug, the start and resolution dates of the episode, and its reported cause. Drug-level shortage spells are defined by molecule and dosage form, such as injectable cisplatin; detailed availability reports also identify which labelers, the firms marketing each version of a drug, are unable to supply it. The records do not, however, measure the quantity of unmet demand (how much supply is missing). The model of Section 4 defines shortages formally, as delivery shortfalls relative to contracted demand, which allows me to quantify their severity.

Key Features of Shortages. Figure 1 displays the yearly number of shortages in these records, distinguishing newly reported shortages from ongoing ones. Shortages often persist for several quarters and can recur for the same drug, so that ongoing episodes accumulate even when fewer new shortages are reported. The reported causes point primarily to problems in production. Among episodes with an identified cause, about two thirds are attributed to manufacturing-quality problems, whereas only 3 percent are attributed to increases in demand (Appendix Figure D.3).[17] Quality problems include chemical or bacterial contamination, sterility failures, and equipment failures that require production

[16] The UUDIS compiles and verifies reports of drug unavailability submitted to the FDA and to the American Society of Health-System Pharmacists (ASHP) by manufacturers and health care providers.

[17] Reporting a cause is not compulsory, and 40 percent of shortage events have no reported cause. Section 3.1 shows that foreign suppliers disappear from U.S. import records when a shortage begins.

to stop or products to be recalled. These interruptions can delay treatment and require less suitable alternatives.[18]

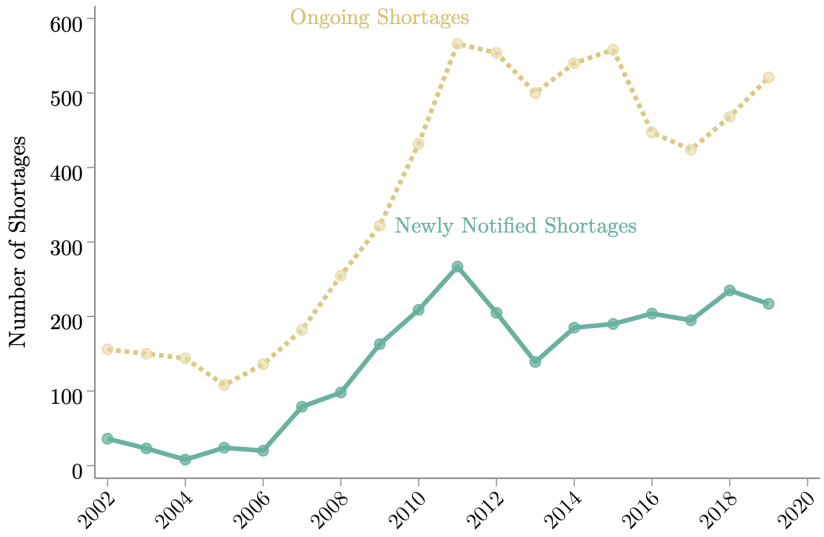


Figure 1: **Yearly U.S. Drug Shortages**

Notes: The figure displays the yearly number of U.S. drug shortages as reported to the UUDIS. A drug is defined at the molecule-form level in the data (e.g., injectable cisplatin). The green, solid line reports the number of new drugs being notified as short in a given year. The yellow, dotted line reports ongoing shortage events, thus capturing the length of disruptions in time.

A production disruption need not create a shortage if existing buffers or alternative suppliers can meet demand. Understanding persistent shortages therefore requires examining both production risk and the market's ability to absorb disruptions.

2.2. Market and Information Structure

Figure 2 summarizes the market structure. Buyers in the United States are hospital groups and their group purchasing organizations (GPOs), which pool purchases and negotiate with drug labelers (generic sponsors) on behalf of member hospitals.[19] Labelers can manufacture drugs in their own plants or outsource production to contract manufacturing organizations. In either case, the plant can be domestic or foreign. The FDA reviews drug applications and oversees manufacturing facilities. Generic sponsors (labelers) obtain marketing approval through an Abbreviated New Drug Application (ANDA), which establishes equivalence to the reference drug and compliance with manufacturing standards (FDA, 2016). Throughout the paper, I distinguish the labeler from the plant producing its drug and measure offshoring by production location, not corporate ownership.[20]

[18]The average shortage affects at least half a million consumers (ASPE, 2023), and the 2011 shortage of norepinephrine increased in-hospital deaths of patients in septic shock by 3.7 percentage points (Vail et al., 2017). Shortages also increase the risk of medication errors and impose costs on hospitals through searches for alternative supplies and changes in treatment

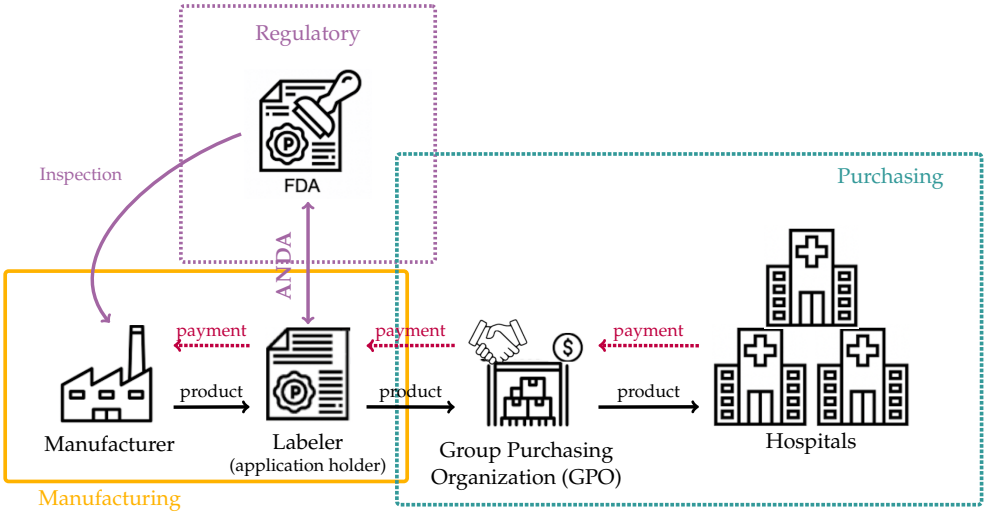


Figure 2: **Market Structure of Generic Injectable Drugs: Manufacturing, Purchasing and Regulation**

Notes: Hospitals buy drugs through wholesalers at prices negotiated by their group purchasing organization (GPO), which selects suppliers through a procurement auction and signs a contract with the labeler, the company holding the marketing license (ANDA) for the drug. The labeler manufactures the drug in a plant it owns or contracts, located in the U.S. or abroad. The FDA delivers the marketing license to the labeler and inspects the plant. Appendix D.1 details the procurement auction.

The U.S. Procurement System. In practice, GPOs solicit bids from eligible labelers and award contracts through competitive procurement auctions (requests for proposals).[21] Hospitals purchase the drugs through wholesalers at the GPO-negotiated prices. Sole sourcing is common within individual contracts, but different hospital groups can select different suppliers of the same drug. A labeler's national sales therefore reflect purchases across multiple buyers, rather than the outcome of a single national tender. Appendix D.1 describes the procurement auction.

Short-run Price and Capacity Rigidities. The standard procurement contract lasts one to two years, during which suppliers are expected to honor the negotiated price. Prices are therefore determined before disruptions materialize and cannot freely adjust within existing contracts.[22] Prices can change when contracts are renewed, and hospitals may purchase outside existing agreements, but these possibilities do not provide immediate adjustment of all contracted purchases. Appendix D.3 documents price movements around shortages using historical list prices.

Production capacity is also slow to adjust, as sterile injectables require specialized equipment and tightly controlled production environments. Changing the products manufactured on a line or adding

protocols (FDA, 2019a).

[19]Between 96 and 98 percent of hospitals buy drugs through GPOs, and 80 percent of hospital purchases go through them. As of 2018, the four largest GPOs accounted for 90 percent of the medical supply market.

[20]The model treats a supplier as a labeler together with the plant it uses to serve a given molecule-form market. This abstracts from the contractual relationship between a labeler and its manufacturing contractor while retaining the plant's production location.

[21]The GPO evaluates proposals and negotiates contract terms before awarding a contract. Price is a central criterion among suppliers holding the required marketing authorization.

[22]In an interview I conducted on February 22, 2022, Todd Ebert, then president of the Healthcare Supply Chain Association, emphasized that GPOs work with suppliers to have contracts honored at the negotiated price.

new lines is costly and slow. Manufacturers allocate capacity in anticipation of procurement demand; when one supplier is disrupted, other plants may already be committed to producing other drugs. Capacity to replace lost output must therefore generally be arranged in advance.[23] Sterile injectables can require refrigeration or protection from light. These storage requirements, together with limited shelf lives, make it harder for hospital pharmacists to stockpile these medicines in anticipation of shortages (Haninger, Jessup, and Koehler, 2011).[24] Clinical interchangeability alone cannot ensure that another supplier has the capacity to deliver the drug.[25]

Suppliers lose revenue when they cannot deliver, but contracts provide limited additional compensation for shortages. Maintaining spare capacity is costly even when it is unused, whereas the costs of delayed or foregone treatment are borne by hospitals and patients. The private return to reliable supply therefore need not reflect its full value to buyers (Appendix D.1).[26]

Institutionalization of Therapeutic Interchangeability. The regulatory framework established by the Hatch–Waxman Act institutionalized therapeutic interchangeability among approved generic versions of a drug.[27] Patients and hospital pharmacists generally do not select the generic manufacturer; purchasing intermediaries choose suppliers on their behalf. Conditional on regulatory approval, suppliers compete primarily on price. Yet this common assurance of therapeutic equivalence does not distinguish suppliers by their ability to deliver consistently. When purchasers cannot readily observe differences in manufacturing reliability, investments that improve supply continuity need not command a corresponding price premium.[28] Current contracting practices thus contribute to a "race to the bottom" in pricing (FDA, 2019a).

[23] Only about one percent of generic sterile injectable applications approved between 2000 and 2011 listed more than one facility for production of the finished drug, compared with almost 20 percent of branded applications (Woodcock and Wosińska, 2013).

[24] Hospital stocks can resolve brief interruptions, but do not add production capacity, and their usefulness is limited by expiration and storage requirements. Shortage spells last well over a year on average (Section 3.1), so that a stock large enough to fully resolve a shortage would often expire before it is used. The model in Section 4 thus represents insurance against disruptions through manufacturers' choices of capacity buffers rather than through buyers' inventories.

[25] Limited substitution between different treatments is distinct from buyers' choice among suppliers of the same molecule and dosage form. The model allows the latter choice to depend on price and supplier characteristics.

[26] Failure-to-supply clauses exist in some contracts, but they reimburse only the extra cost of buying from another supplier, so they do not compensate buyers when no alternative supply exists; moreover, the Uniform Commercial Code excuses a seller from delivering when an unforeseen event makes delivery commercially impracticable (U.C.C. §2-615). In practice, these clauses are rarely invoked (Appendix D.1).

[27] For injectable solutions that contain the same active and inactive ingredients in the same concentration as the brand, the FDA waives the bioequivalence study (21 C.F.R. §320.22(b)(1)).

[28] In congressional testimony, the FDA states that "This lack of transparency reinforces competition based solely on price" and links it to weak incentives to upgrade facilities and quality practices (FDA, 2019b). Reimbursement can reinforce this incentive: when it does not depend on which generic supplier is chosen, a lower acquisition price increases the hospital's margin (Yurukoglu, Liebman, and Ridley, 2017; Conti and Wosińska, 2025). In retail generic drug procurement, Ishitani (2025) finds that purchases fall by 60 percent following recalls but respond little to failed plant inspections. Recalls are publicly announced and documented for each product, whereas manufacturers are not required to report the cause of a shortage.

Regulatory Oversight. The FDA monitors compliance with Current Good Manufacturing Practices through plant inspections. Domestic and foreign facilities face the same manufacturing standards, but the frequency of inspections differs across locations. From FDA inspection records over 2004–2019, I calculate frequencies corresponding to about one inspection every 2.5 years in the United States, every 5 years in other high-income countries, and every 14 years in India and China. In 2016, an estimated 33 percent of foreign facilities had never been inspected (GAO, 2016a). Overseas inspections also face logistical constraints and are more often announced in advance (FDA, 2019b).[29] These differences in monitoring accompany the production-cost advantages that can make offshore plants attractive to labelers. Additional Results Section S1.3 describes the inspection records and their interpretation.

What Buyers Observe. Buyers observe the bid price, labeler, and marketed product, but generally not the identity of its manufacturing plant, its outsourcing arrangements, or its investments in redundant capacity.[30] Procurement occurs at the labeler level, while production risk arises at the plant. My data recover that plant; the buyer usually cannot. Past shortages provide information about a labeler's deliveries without fully revealing these underlying arrangements: the plant behind a labeler may change, and the labeler may supply different purchasing organizations over time. Thus, past purchasing relationships are an imperfect guide to future plant-level risk.[31] Additional Results Section S1.1 discusses how decentralized purchasing can further weaken incentives to pay for shared production capacity.

These procurement and technological frictions predate the shift in manufacturing abroad. Offshoring changes the set of plants supplying the market without necessarily changing how procurement rewards their reliability. The model of Section 4 examines this interaction: manufacturers choose location, price, and capacity before supply shocks occur, while prices, capacity, and buyer allocations remain fixed within the contract period.

2.3. Mapping Drug Products to Manufacturing Plants

I construct a panel dataset linking U.S. injectable drug products to their labelers, manufacturing plants, prices, quantities, and shortage histories. Existing sales data identify the firms marketing drugs, while facility records locate plants. Neither alone provides a history of which plants supply each product.[32] A labeler's headquarters can be far from its manufacturing sites, and aggregate trade data do not identify individual drugs. I combine product and plant records with detailed shipment data to reconstruct these

[29] In India and China, FDA inspectors need to announce visits in advance to obtain visas, requiring the foreign company's sponsorship and the state regulator's approval. The inspection-frequency measure enters the model as a regulatory cost shifter (Section 4.3.2).

[30] The product's National Drug Code does not identify the plant either. One code can be filled at several facilities, and a labeler can add or switch manufacturing sites through a supplement to its approved application without changing the code, since a new code is required only when the product itself changes (21 C.F.R. §§ 207.35, 314.70 and 314.97).

[31] Appendix D.4 shows that labelers whose product went short keep selling it at nearly the baseline rate, in sales aggregated over all U.S. buyers: of the labelers selling a product this year, 99.1 percent still sell it next year when the product delivered and 98.2 percent when it went short, and their market share does not decline the year after a shortage.

[32] The difficulty of establishing these links was confirmed in a discussion with FDA economists from the Drug Shortage Team in 2020.

links. Appendix F.1 describes each data source; Figure F.1 summarizes the data structure, built around three main blocks.

The first block describes the downstream market. I combine the FDA's NDC Directory and Orange Book to identify injectable products and their characteristics, including their labelers, dosage forms, approval dates, and generic or brand status.[33] These sources include drugs that never appear on a shortage list as well as drugs that do. I then link products to the UUDIS shortage histories. Beyond the drug-level episodes, I digitize thousands of detailed availability documents to extract the affected labeler-products, their National Drug Codes (NDCs), shortage dates, and reported reasons. This identifies which suppliers are unable to deliver within a drug market.

I complement these data with quarterly sales revenue and physical quantities for each product from IQVIA, aggregated across all U.S. buyers.[34] I compute average realized unit prices by dividing revenue by the number of units sold. For descriptive exercises, I also use list prices from the Red Book (IBM Micromedex), a proprietary database: its historical pricing tables, complemented with the active price records I collected from it. The sales-based prices used in estimation come from IQVIA.

The second block identifies manufacturing plants. Registration records from the FDA's Center for Drug Evaluation and Research (CDER) provide establishment names, addresses, identifiers, and annual registrations for domestic and foreign facilities. I complement these records with facility lists collected under the Generic Drug User Fee Amendments (GDUFA) and with FDA inspections and warning letters. These data locate establishments, but registration alone does not establish which drugs they actually produce or their output volumes.

The third block links the first two using detailed customs records from Descartes Datamyne: the bills of lading of U.S. maritime imports, U.S. export records, and the export records of foreign countries, mainly India, which include their shipments to the United States. I use text analysis to identify drug names and dosage forms in detailed shipment descriptions. The shipper's name and address identify the exporting entity; matching that address to an FDA establishment links the drug to a manufacturing site rather than merely to a corporate name. I standardize and geocode names and addresses in both sources, match records using text similarity and geographic proximity, and review ambiguous matches manually. U.S. import manifests identify foreign suppliers, supplemented by Indian export records. U.S. export records also link drugs to domestic plants, which need not appear in U.S. import data. Appendix F.2 describes the matching procedure and record linkage techniques.

The resulting data distinguish registered product–plant relationships from observed shipments. The annual set of registered plants is used to measure a drug's production geography; shipment records show which foreign suppliers send the drug to the United States in a given period. A supplier's absence from shipments during a shortage therefore does not itself remove that supplier from the registered-plant measure.

I link 95 percent of the injectable products on the shortage list to at least one manufacturing plant

[33] The NDC Directory lists every drug product marketed in the U.S. under its National Drug Code; the Orange Book lists FDA-approved drugs with their approval dates and generic or brand status.

[34] I do not see hospitals' (or GPOs') direct purchases. On the demand side, I only have quarterly data at the NDC 5-4-2 level (a labeler-molecule-dosage form-package level) for the entire U.S. market, through IQVIA. Prices I use are thus average prices over all U.S. hospitals for a given NDC 5-4-2 product.

(Appendix F.2). The data record realized sales, not the quantities buyers would have purchased without a supply constraint; I estimate this unmet demand in Section 5. The next section uses this dataset to assess whether increases in shortage frequencies and lengths are disproportionately driven by disruptions in manufacturing facilities that are foreign.

3. Reduced-Form Evidence: Foreign Production and Shortages

I use the dataset described in Section 2 to document new empirical evidence on (i) the offshoring of drug manufacturing and (ii) the relationship between foreign-made products and an increased likelihood and intensity of shortage spells. Subsection 3.1 documents three stylized facts that link the two phenomena: foreign suppliers disappear from U.S. import records when a shortage begins; drugs that go short are more offshored and imported drugs face more frequent and longer shortages; and drugs whose demand grows in Asia are the ones pulled abroad, and in turn the ones in shortage in the U.S. Offshoring is, however, an endogenous decision of the firm, and there is selection of the type of drugs that go abroad. Subsection 3.2 addresses this selection by following each drug across the year its production moves abroad and comparing its shortage outcomes with those of drugs whose production stays in the United States, in a dynamic difference-in-differences design.

3.1. Stylized Facts on Foreign Manufacturers and Shortages

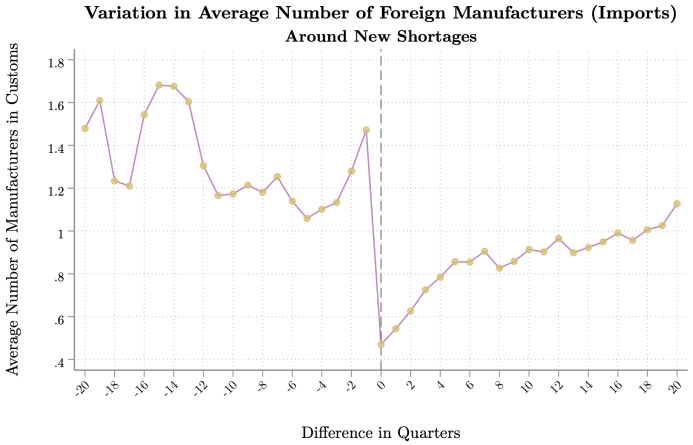
Definition of foreign production and offshoring. Throughout the empirics, I measure foreign production at the molecule–form level by a drug's *share of foreign manufacturers*, defined as the share of the registered manufacturing plants for a drug that are located outside of the U.S. in a given year.[35] In the event-study analysis below (Section 3.2), I additionally define an *offshoring event* when a molecule–form transitions from being produced only in U.S. plants to having any registered foreign plant; the corresponding indicator turns on in the year of that transition. My analysis is agnostic about ownership. A molecule–form can be "offshored" either because a U.S. labeler (generic sponsor) moves production from a domestic plant to a foreign facility, or because a new foreign labeler enters and supplies the U.S. market through its foreign plants. In both cases, the relevant object for shortages is the actual location of production, regardless of whether the plant is vertically integrated or contracted out to a contract manufacturing organization (CMO).

Fact 1: Foreign imports drop during shortages. Figure 3a reports quarterly variation in the average number of foreign manufacturers of a drug (defined at the molecule–form level) around the occurrence of a shortage, as observed in U.S. import manifests (bill-of-lading). The start of a shortage coincides with a sharp drop in the number of foreign manufacturers visible in U.S. import records, which then remains depressed for several quarters. This pattern is informative for two reasons. First, it validates

[35] Plants are identified from the FDA facility registration data and linked to drug products in customs and shortage records using probabilistic record linkage on firm names and geo-located addresses (Appendix F.2). Customs records are the bill-of-lading data on U.S. imports (Descartes Datamyne), complemented by U.S. and Indian export records, which list the shipper and the product of each shipment.

the customs-based linkage between drugs and foreign plants: import activity for a drug moves in tandem with its shortage status. Second, it supports that foreign suppliers are directly exposed to the underlying disruptions: they stop or sharply reduce shipments to the U.S. when disruptions occur, thus temporarily disappearing from customs records at the onset of shortages.[36]

(a) Foreign importers drop during shortages



(b) Imported drugs face more frequent and longer shortages

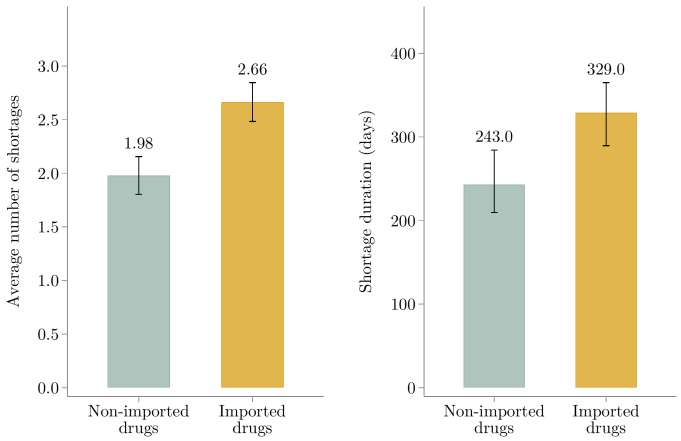


Figure 3: **Correlations between Offshoring and Injectable Shortages**

Notes: Panel (a): This figure reports an event study of the average number of foreign manufacturers captured in the U.S. import customs data for sterile injectable drugs that experience at least one shortage. For each drug, defined at the molecule-form level, I compute the number of distinct foreign manufacturers by quarter in the customs data and then average across drugs (the granularity of the bill-of-lading data is at the daily level). Quarter 0 corresponds to the quarter in which a new shortage is first reported. A shortage event coincides with a sharp drop in the number of foreign manufacturers visible in import records. Panel (b): for the sterile injectable drugs reported in shortage between 2004 and 2019, average number of shortage spells per drug and median duration of a spell, with 95 percent confidence intervals, for drugs with at least one foreign manufacturer found in U.S. import customs records (imported) and for the others. Spells still open at the end of 2019 are right-censored at their last observed month.

The share of foreign manufacturers used in the next two facts comes from the comprehensive matching of drugs to their *registered* plants.[37] Compared to the import records of Fact 1, this matching ensures that the presence of firms in the data is not affected by the occurrence of shortages: for each year, I capture the *exact set* of firms registered to manufacture injectable drugs for the U.S. market.

Fact 2: Injectable drugs experiencing shortages are more offshored than sterile injectables that do not, and imported drugs face more frequent and longer shortages. In Appendix Figure E.1 I look at yearly variations in the share of foreign manufacturers for injectable drugs that went into shortage *at least once* over the 2004-2019 period and compare it to injectable drugs that did not go short. Sterile injectable drugs experiencing shortages display higher shares of foreign-based manufacturers compared to sterile injectables that never went into shortages, and this share also increases more over time: from

[36] For each molecule–form, I count the number of distinct foreign manufacturers appearing in U.S. bill-of-lading data in each quarter and then average this count across drugs. This descriptive fact relies on the sole merge between the drug shortages list and import manifests (bill-of-lading data); because the customs data record realized shipments, this step alone does not recover the full set of FDA-registered plants authorized to supply a drug in a given year.

[37] It adds to the previous data of drug products and shipments a second matching between the U.S. customs and the FDA's dataset of manufacturing plant locations. This matching was made using probabilistic record linkages, both exploiting firms' names and their geo-located addresses in the U.S. Customs and in the FDA data.

23% to 67% for drugs short, against 4.3% to 33% at its 2017 peak for non-short drugs. Imported drugs also face more frequent and longer shortage spells than non-imported drugs: among sterile injectables reported in shortage between 2004 and 2019, drugs supplied by foreign manufacturers are reported in shortage more often, 2.7 times on average over the period against 2.0 for drugs with no foreign manufacturer identified in U.S. import records, and their shortage spells last longer: 329 days at the median against 243, and 617 days on average against 487 (Figure 3b).[38] Taken together, these patterns indicate that sterile injectables tied to foreign plants are both more exposed to shortages and subject to longer disruptions than drugs with no foreign manufacturer identified in those records.

Fact 3: Manufacturers are disproportionately likely to offshore manufacturing to Asia for drugs that are in increasing demand there as their populations age, and these drugs are in turn the most likely to be reported in shortage in the U.S. I establish this fact with a shift-share instrument for offshoring built on the home-market effect (Krugman, 1980): as the demand for a drug grows in a country, manufacturers locate its production closer to that market. The instrument is a predicted disease burden, the burden expected in India and China given their demographic structure and insurance coverage, for the diseases a drug treats (Acemoglu and Linn, 2004; Costinot et al., 2019); Appendix E.3 details its construction and reports the regressions.[39] All else equal, moving the production of a drug from entirely domestic to entirely Asian manufacturers raises the probability that a new shortage occurs in a given year by 30 percentage points, against a yearly baseline of 7.7 percent. It also raises the fraction of the year during which the drug is in shortage by 95 percentage points, that is, about 350 additional days of shortage a year for a drug short 44 days a year on average. For foreign production as a whole, the effects are 24 and 74 percentage points.[40] The instrument thus identifies which drugs are offshored because of the growth of demand in Asia, and shows that these are the drugs that go short. What it does not show is when their production moves and how their shortages evolve afterwards, which is the object of the next subsection.

3.2. Offshoring and Shortages: Dynamic Difference-in-Differences

Post-offshoring, injectable drugs are more likely to experience shortages than similar drugs whose production remained in the U.S. Drugs whose production leaves the United States are not a random sample of the market, and a manufacturer who expects a product to become unprofitable may both move it abroad and stop investing in its supply. To address selection into offshoring, I follow each

[38] I define an imported drug as a sterile injectable for which at least one foreign manufacturer is found in the U.S. import customs records over the period. All statistics are computed over the sterile injectable drugs reported in shortage between 2004 and 2019. One spell in ten is still open at the end of 2019, more often for imported drugs, so that the duration gap is, if anything, understated. Appendix E.1 reports the underlying statistics.

[39] The age structure of India and China and the disease rates of the other countries that enter the index do not depend on where a drug is produced, and they matter for the supply of U.S. hospitals only through that location: a surge in demand in India could divert supply away from the U.S. only once the drug is produced there. Appendix E.3 discusses this channel.

[40] Each additional ten percentage points in a drug's share of Asian manufacturers raises the probability of a new shortage by 3 percentage points and the time in shortage by 9.5 percentage points of the year, about 35 additional days; for the share of foreign manufacturers the figures are 2.4 and 7.4 percentage points, about 27 days. Among the drugs that go short at least once over 2004 to 2019, the average time in shortage is 77 days a year.

sterile injectable through the year its production first moves abroad and compare its shortage outcomes with those of drugs whose production remains in the U.S. Offshored drugs become about 9 percentage points more likely to be in shortage, against a mean of roughly one year in four, and the days they spend in shortage rise by about three quarters.

The unit of observation is a drug, a molecule and dosage form, followed annually from 2002 to 2019. An offshoring event is the first year in which a drug has a registered manufacturing plant outside the United States, and a drug remains offshored thereafter.[41] Of the 480 drugs in the panel, 299 are offshored during the sample period and 65 are verified domestic throughout, with every registered plant in the United States.[42] Because offshoring events are staggered, I estimate the group-time average treatment effect of Callaway and Sant'Anna (2021). For the cohort of drugs offshored in year g and a post-event year $t \geq g$,

$$\mathrm{ATT}(g,t) = \mathbb{E}\left[\mathrm{Shortage}_{jt} - \mathrm{Shortage}_{j,g-1} \mid G_j = g\right] - \mathbb{E}\left[\mathrm{Shortage}_{jt} - \mathrm{Shortage}_{j,g-1} \mid G_j > t\right], \quad (1)$$

where $\mathrm{Shortage}_{jt}$ is the shortage outcome of drug j in year t and G_j is the year in which drug j first registers a foreign plant, with $G_j = \infty$ for drugs that never do, so that the control group in year t is the drugs not yet offshored, the verified-domestic drugs included. Drugs that will move later share whatever makes a drug a candidate for offshoring, which is why they are the comparison of choice; the 65 drugs that never move are, by construction, the drugs no manufacturer ever chose to relocate. The identifying assumption is that, had its production stayed, an offshored drug's shortage record would have moved like that of the drugs still produced domestically; the pre-event coefficients test it. Before the event the base year is the preceding year, so that pre-event coefficients are year-on-year differences between the two groups rather than normalizations. The event-study coefficients β_k plotted in Figure 4 average these effects across cohorts at the same distance k from the event, weighting each cohort by its size, and the average effect reported in the tables averages $\mathrm{ATT}(g,t)$ over the event year and the ten years after it.[43]

Figure 4 plots the β_k for the probability of shortage and for the fraction of the year a drug spends in shortage. Three main patterns emerge. First, the five pre-event coefficients, each the difference between the two groups in the year-on-year change of the outcome, are small, under 2 percentage points on average, and jointly indistinguishable from zero (Appendix Table E.4). Second, after the event the probability of being in shortage rises and remains elevated. It is close to zero for the first three years, builds from the fourth, and reaches about 14 percentage points by the fifth year and about 20 from the eighth. The build-up is consistent with how production moves in this market: a first foreign plant rarely replaces domestic lines at once, and the domestic capacity that absorbed disruptions exits over

[41] Plants are linked to drugs through the record linkage of Appendix F.2, which applies to domestic and foreign sites alike. I date the event on the first evidence of a foreign plant rather than on the share of a drug's NDCs produced abroad, which tilts toward foreign sites because plants enter the customs-based records only when they ship; the move itself may predate the first year in which it is observed.

[42] A further 25 drugs are already produced abroad in 2002 and 91 have no usable plant record; neither group enters the estimation.

[43] No covariates enter the estimates, and standard errors are clustered by drug. Ten years is the last horizon at which more than two fifths of the offshored drugs are still observed. Appendix E.2 details the design.

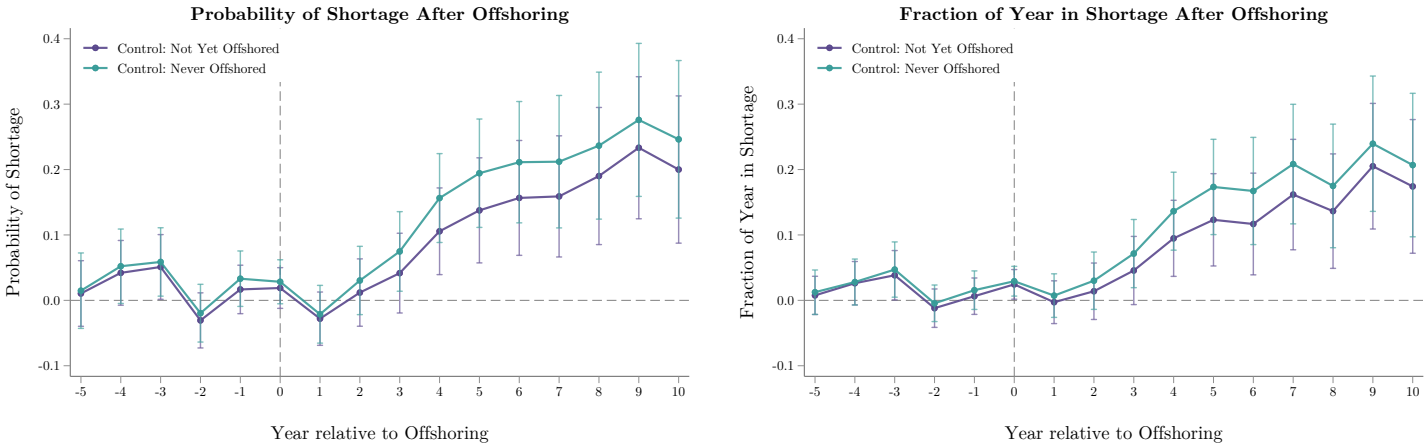


Figure 4: **Event Studies: Effect of Offshoring on Shortage Spells**

Notes: Each panel plots the event-study coefficients $\beta_k = \sum_g w_{gk}\,\mathrm{ATT}(g, g+k)$, $k = -5, \ldots, 10$, which aggregate equation (1) by year relative to the offshoring event with weights w_{gk} proportional to the number of drugs of cohort g observed at horizon k, with 95 percent confidence bars from standard errors clustered by drug, on the 299 offshored and 65 verified-domestic drugs. The left panel uses an indicator equal to one if the drug is in shortage; the right panel the fraction of the year during which at least one of the drug's NDCs is in shortage. The two lines differ only in the control group: the purple line compares offshored drugs with drugs not yet offshored in that year together with the verified-domestic drugs, the comparison used throughout; the green line compares them with the verified-domestic drugs alone, which never offshore.

the following years. Averaged over the event year and the ten years after it, offshored drugs are about 9 percentage points more likely to be in shortage than drugs not yet offshored, against an average of 27 percent among the drugs that move, and about 7 points more likely to have a new shortage notified (Appendix Table E.3). Third, offshored drugs spend longer in shortage. The days a drug spends in shortage, counted from a shortage's first notification to its resolution and zero in years without one, rise by about three quarters against drugs not yet offshored and more than double against never-offshored drugs, roughly 70 and 120 additional days.[44] Offshored drugs also spend about 30 more days of each year in shortage, over a third more than the 80 days a year among drugs that move. Against never-offshored drugs alone, the green lines in Figure 4, every effect is larger, so the comparison used throughout is the smaller of the two. The results are also robust to the definition of the control group (Appendix Figure E.2) and to the choice of estimator (Appendix Figure E.3).

The two designs rest on different variation. The instrument compares drugs whose demand grows faster in Asia with drugs whose demand grows less; the event study compares the change in a drug's shortages after its production moves abroad with the change, over the same years, among drugs still produced domestically. Under their respective identifying assumptions, both designs indicate that offshoring increases shortage incidence and duration. In the context of the model in Section 4, these patterns are consistent with manufacturers shifting production toward plants with higher disruption risk and limited ex-post scope to expand capacity or adjust prices, so that offshoring translates into

[44] Days in shortage are heavy tailed and zero in most drug-years, so I estimate the effect on the logarithm of one plus the days (0.569 and 0.836 log points). The log effect combines more frequent and longer shortages, which is why the level estimates, 69 and 121 days against an average of about 300 among the drugs that move, are smaller in proportion; only the second is distinguishable from zero (Appendix Table E.5).

persistently higher shortage risk. The estimates do not reveal why manufacturers offshore the drugs whose supply then deteriorates.[45]

3.3. Summary of Empirical Findings

This section presented novel empirical evidence linking foreign manufacturers to drug shortage spells. First, a shortage in the U.S. coincides with a drop in imports of the affected drug products. Second, injectable drugs that go short are more offshored than those that do not, and drugs whose manufacturers are located overseas suffer longer and more frequent shortages. Third, using variation in predicted disease burden, I find that manufacturers are disproportionately likely to offshore manufacturing to Asia for drugs that are in increasing demand there as their populations age, and that these drugs are in turn the most likely to be reported in shortage in the U.S. Finally, following each drug across the year its production moves abroad, I find that offshoring causes shortages: offshored drugs become about 9 percentage points more likely to be in shortage, and the time they spend in shortage rises by about three quarters.

Taken together, these facts show that foreign plants, and plants in the Global South in particular, are systematically riskier, and that moving the production of a drug abroad makes it go short more often and for longer. What, then, motivates manufacturers to let supply fail, and why is offshoring linked to persistent rather than transitory shortages? The reduced-form evidence cannot answer these questions. It establishes the link between offshoring and shortages, but the estimates measure how often and for how long drugs go short, not how much supply is missing, nor what it would cost to restore it. Answering these questions requires a model of the incentives behind manufacturers' location and production choices, and of the market frictions through which a disruption at a plant becomes a shortage in the U.S. The model in Section 4 formalizes these incentives; its estimation quantifies unmet demand during shortages and the price and welfare effects of alternative location policies and procurement contracts.

4. A Model of Drug Shortages in Global Markets

This section develops a partial-equilibrium model of global procurement for injectable drugs that generates shortages endogenously. Shortages arise when upstream supply shocks interact with two short-run rigidities: fixed-price procurement contracts, which limit price adjustment within the contracting period, and lumpy capacity for sterile injectable lines, which prevents short-run capacity response to disruptions. Manufacturers choose production locations, bid prices, and production capacity before location-specific supply shocks are realized. Capacity buffers are therefore costly ex ante investments rather than an ex post adjustment margin. A shortage occurs when realized supply falls short of contracted quantities.

[45] The estimates net out everything that is fixed about a drug over time, but they do not rule out that manufacturers offshore precisely the drugs whose supply is about to deteriorate; the model returns to this selection by treating the location decision itself as a choice.

Section 3 leaves open why manufacturers offshore production despite the associated shortage risk. In the model, offshoring is a lever for cost-cutting: firms can choose lower-cost locations with higher disruption risk, while procurement rewards lower bids but not plant-level reliability directly. Manufacturers bear the revenue losses from unfulfilled contracts, but not the resulting losses to buyers. I estimate the model and use it to compare counterfactual contract designs and location policies in terms of prices, drug availability, and welfare.

4.1. Environment and timing

Environment. A market m consists of competing injectable drug products, and period t is a calendar year. Buyers are U.S. hospitals and their group purchasing organizations, each acting as a monopsonist for its set of patients (Dafny, 2005).[46] Suppliers compete in prices for procurement contracts; each supplier is a labeler together with the plant it uses to serve the market. I take the set of approved suppliers J_{mt} as given and abstract from entry and exit, as in Dubois and Lasio (2018).[47]

Suppliers choose locations, bid prices, and capacity before supply shocks occur, and buyers allocate contracted quantities at these prices. Prices, capacity, and buyer allocations then remain fixed within the period.[48] This timing represents the contracting and production rigidities described in Section 2.2: prices cannot freely adjust within existing contracts, and sterile production lines cannot be expanded or reassigned immediately after a disruption.

Timing within period t.

1. **Location choice.** Each manufacturer $j \in J_{mt}$ chooses the location $\ell_{jmt} \in \{H, F_N, F_S\}$ of the production plant serving market m in period t. The location determines the manufacturer's marginal-cost primitives and the distribution of its capacity-yield shock.
2. **Bid and production capacity choice.** Manufacturers j choose bid prices p_{jmt} and commit period production capacity κ_{jmt} for each market m.
3. **Tenders.** U.S. buyers allocate contracted quantities Q^D_{jmt}. Prices and capacity cannot adjust within the contracting period t.
4. **Shock realization and shortages.** After location, pricing, and production capacity decisions are made by firms, a supplier-level capacity yield $\omega_{jmt} \sim F_{\ell_{jmt}}$ realizes. Manufacturer j's realized market supply is $q^{\text{real}}_{jmt} = \min\{Q^D_{jmt}, \kappa_{jmt}\omega_{jmt}\}$.

[46] There is no competition at the patient level, and hospitals typically fully internalize the prices of drugs they purchase on behalf of patients. Similar to Dubois and Lasio (2018), I cannot observe data on the behavior of insurers, healthcare providers, and other intermediaries between patients and manufacturers. Hence, I abstract away from modeling them and do not disentangle their role in aggregate revealed preferences.

[47] Institutional note: entry decisions take place at the time manufacturers apply for a marketing license (an "ANDA"). Sunk entry and certification costs are thus only paid once, prior to the start of the procurement game. Conditional on holding an ANDA, the administrative cost of submitting a bid in a tender is negligible. ANDAs do not have expiration dates.

[48] Contract prices are typically firm-fixed for the one- to two-year term. Prices are typically only renegotiated after contracts expire and a new procurement auction takes place. Many contracts include "price protection" clauses capping mid-contract increases; these are private contractual terms, not regulatory caps.

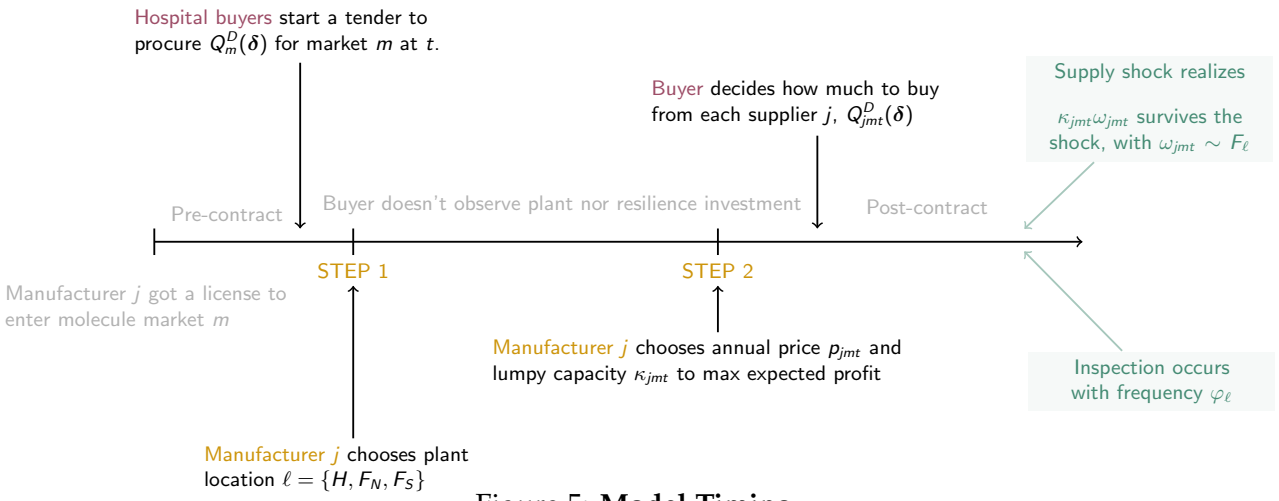


Figure 5: **Model Timing**

Notes: A downstream hospital buyer located in a single location ('Home') starts a tendering process to procure a given quantity $Q_{mt}^D(.)$ in market m at time t. The given set of J_{mt} approved manufacturers competes each period for procurement contracts. Each manufacturer j's problem is decomposed into two steps. In a first step, each manufacturer $j \in J_{mt}$ chooses the location of the production plant serving market (m,t). In a second step, they choose production capacities κ_{jmt} and unit prices p_{jmt} to maximize expected profit. Products are differentiated: buyers assign market shares based on manufacturers' proposed prices and observable characteristics. After contracts are assigned, a supplier-level capacity-yield shock $\omega_{jmt} \sim F_{\ell_{jmt}}$ hits and may reduce realized market supply relative to contracted demand. Manufacturers have two margins to insure against supply shocks: first, building capacity buffers; second, choosing production locations with lower disruption risk.

4.2. Buyer's Demand for Drugs

A Discrete Choice Logit Model. Each period t, buyers choose among J_{mt} differentiated injectable drug products and an outside option.[49]

Buyer i's indirect utility from supplier j in market m at time t is

$$v_{ijmt} = \underbrace{-\alpha \log p_{jmt} + \boldsymbol{\beta}' \mathbf{X}_{jmt} + \xi_{jmt}}_{\equiv \delta_{jmt}} + \varepsilon_{ijmt}, \tag{2}$$

with the mean utility of the outside option (no purchase) normalized to zero:

$$v_{i0mt} = \delta_{0mt} + \varepsilon_{i0mt} = \varepsilon_{i0mt}, \qquad \delta_{0mt} = 0.$$

Manufacturer j-specific observable characteristics are captured by the bid price p_{jmt} and by a vector $\mathbf{X}_{jmt} = (g_{jm}, A_{jmt})$, where g_{jm} is a categorical brand-status variable taking values in $\{\text{brand}, \text{branded generic}, \text{generic}\}$, and A_{jmt} is product-line breadth, measured as the number of distinct package-size presentations offered by supplier-product j in market m and year t. The term ξ_{jmt} captures unobserved heterogeneity at the supplier–market–year level. The parameter $\alpha > 0$ governs buyers' disutility from higher prices.[50] The idiosyncratic component ε_{ijmt} is i.i.d. type I extreme value

[49] Appendix D.4 shows that suppliers continue selling after shortages at nearly the baseline rate, and I find no evidence of a subsequent decline in their national market shares once supply has recovered. These results concern aggregate purchases across U.S. buyers and motivate omitting shortage history from the baseline demand specification.

[50] Note that because price enters in logs, α is not a constant marginal utility of income. A log price specification fits the data better, due to the large heterogeneity in prices across markets. Related work has also adopted this specification for the same reasons (Atal, Cuesta, and Sæthre, 2026; Dubois and Lasio, 2018).

(T1EV), and δ_{jmt} represents the mean utility.

The T1EV distribution of the logit error term yields the standard mapping from indirect utility to model–predicted market shares and contracted quantities:

$$s^D_{jmt} = \frac{\exp\{-\alpha \log p_{jmt} + \boldsymbol{\beta}' \mathbf{X}_{jmt} + \xi_{jmt}\}}{1 + \sum_{k=1}^{J_{mt}} \exp\{-\alpha \log p_{kmt} + \boldsymbol{\beta}' \mathbf{X}_{kmt} + \xi_{kmt}\}}, \tag{3}$$

$$Q^D_{jmt}(\delta) = M_{mt} s^D_{jmt}, \tag{4}$$

where Q^D_{jmt} denotes demanded quantities (quantities *contracted* at the procurement auction stage) and M_{mt} is the size of market m in period t. Each manufacturer j's market share and demanded quantity depend on the characteristics of all competing manufacturers in the market.

4.3. Technology, Costs, and Shocks

4.3.1. Uncertainty and Shortages

After contract prices p and production capacities κ are chosen, an exogenous supplier-level capacity-yield shock realizes. Conditional on the chosen plant location ℓ_{jmt},

$$\omega_{jmt} \sim F_{\ell_{jmt}}, \qquad \omega_{jmt} \in [0, 1], \tag{5}$$

where F_ℓ denotes the capacity-yield distribution associated with production location ℓ. The shock reduces usable capacity from κ_{jmt} to $\kappa_{jmt}\omega_{jmt}$. The capacity yield ω_{jmt} is a random variable that captures the share of supplier j's capacity that survives the shock and can be used to fulfill contracted demand.[51]

Definition 4.1 (Demand-to-capacity ratio)**.** *Define the* demand-to-capacity ratio *for manufacturer j in market m and period t as*

$$\boxed{\theta_{jmt} \equiv \frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}}. \tag{6}$$

In equilibrium, $\kappa_{jmt} \geq Q^D_{jmt}(\delta)$ always holds; manufacturers choose production capacities to be weakly above contracted quantities Q^D_{jmt} (see Appendix A.1 for a formal proof). Hence $\theta_{jmt} \in (0, 1]$.

Definition 4.2 (Firm-level Shortages)**.** *For a supplier j located in ℓ, a shortage occurs whenever surviving capacity $\kappa_{jmt}\omega_{jmt}$ falls short of contracted demand $Q^D_{jmt}(\delta)$, so that there is positive excess demand:*

$$\kappa_{jmt}\omega_{jmt} < Q^D_{jmt}(\delta) \iff \omega_{jmt} < \theta_{jmt}.$$

The probability that firm j goes short in market (m, t) is therefore

$$\boxed{\Pr(\textit{Shortage})_{jmt} = \Pr\Big(\kappa_{jmt}\omega_{jmt} < Q^D_{jmt}(\delta)\Big) = F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right) = F_\ell(\theta_{jmt}).} \tag{7}$$

The shortage probability is thus a direct function of the location-specific yield distribution F_ℓ and of the demand-to-capacity ratio θ_{jmt}. A disruption does not necessarily translate into a shortage: a supplier goes short only if the shock is large enough relative to its capacity buffer.

[51] Conditional on location, capacity-yield shocks are supplier-level draws. This formulation is widely used in the Operations Management literature; see, for instance, Tang and Kouvelis (2011).

Realized supply and shortages. The realization of shortages affects the quantities effectively supplied in the market. For any realization of the shock ω_{jmt}, realized supply is censored:

$$q_{jmt}^{\text{real}} = \min\left\{Q_{jmt}^{D}(\delta), \kappa_{jmt}\omega_{jmt}\right\} \qquad \text{so} \qquad q_{jmt}^{\text{real}} \leq Q_{jmt}^{D}(\delta). \tag{8}$$

In the absence of shortages, realized supply coincides with latent logit demand, $q_{jmt}^{\text{real}} = Q_{jmt}^{D}(\delta)$, and any surviving capacity above demand is unused. During shortages, realized supply falls below demand. For a given realization of the yield shock, the firm-level quantity of unmet demand is

$$Q_{jmt}^{\text{short}}(\omega) = Q_{jmt}^{D}(\delta) - q_{jmt}^{\text{real}} = \max\left\{Q_{jmt}^{D}(\delta) - \kappa_{jmt}\omega_{jmt}, 0\right\} \in \left[0, Q_{jmt}^{D}(\delta)\right]. \tag{9}$$

Definition 4.3 (Excess Demand)**.** *At the firm-level, the share of excess demand is,*

$$s_{jmt}^{\text{excess}}(\omega) \equiv \frac{Q_{jmt}^{\text{short}}(\omega)}{Q_{jmt}^{D}(\delta)} = \max\left\{1 - \frac{\omega_{jmt}}{\theta_{jmt}}, 0\right\} \in [0,1]. \tag{10}$$

Thus s_{jmt}^{excess} measures the fraction of supplier j's latent contracted demand that is unmet in a given state of the world, which summarizes shortage severity.[52]

Let $\boldsymbol{\omega}_{mt} \equiv (\omega_{kmt})_{k \in J_{mt}}$ denote the vector of realized supply shocks in market (m,t). At the market (m,t) level, the realized total quantity of unmet demand is

$$Q_{mt}^{short}(\boldsymbol{\omega}_{mt}) \equiv \sum_{j \in J_{mt}} Q_{jmt}^{short}(\omega_{jmt}), \tag{11}$$

and the market-level share of excess demand is

$$s_{mt}^{\text{excess}}(\boldsymbol{\omega}_{mt}) \equiv \frac{Q_{mt}^{\text{short}}(\boldsymbol{\omega}_{mt})}{\sum_{j \in J_{mt}} Q_{jmt}^{D}(\delta)} = \frac{\sum_{k \in J_{mt}} \max\left\{Q_{kmt}^{D}(\delta) - \kappa_{kmt}\omega_{kmt}, 0\right\}}{\sum_{j \in J_{mt}} Q_{jmt}^{D}(\delta)}. \qquad \text{(Eq. Market Excess)}$$

This object measures the fraction of total latent inside-goods demand that is unmet in market (m,t). By construction, $s_{mt}^{excess}(\omega) = 0$ when no firm is short and $s_{mt}^{excess}(\omega) \in (0,1]$ when there is unmet demand.

4.3.2. Location-specific factors

A production location ℓ determines three location-specific primitives: (i) the capacity-yield distribution F_ℓ, (ii) the marginal-cost shifters entering c_{jmt}^{ℓ}, and (iii) the common element of fixed costs FC_ℓ.[53] For supplier j serving market (m,t), these primitives are evaluated at its chosen plant location ℓ_{jmt}.

[52] Under the assumption that latent demand flows uniformly within the contract period, $s_{jmt}^{\text{excess}}(\omega)$ is also the model counterpart of the observed fraction of the period during which product j is in shortage. The model therefore distinguishes two empirical objects. The extensive margin is the probability that a shortage occurs, while the intensive margin is the expected firm-level share of excess demand, $\mathbb{E}_{\omega \sim F_\ell}\left[s_{jmt}^{\text{excess}}(\omega)\right] = F_\ell(\theta_{jmt}) - \frac{h_\ell(\theta_{jmt})}{\theta_{jmt}}$. Appendix B.6.1 derives this equation.

[53] I suppress time variation in the common fixed-cost component here. In estimation, this component varies across the groups of years defined in Section 5.3.3.

Marginal costs. Marginal costs are parametrized by:

$$c^{\ell}_{jmt} = \frac{w_{\ell t}\,\tau_{\ell t}}{\gamma_{jmt}}\,\varphi_{\ell}, \tag{Eq. MC}$$

where $w_{\ell t}$ is a location-specific wage in the pharmaceutical manufacturing sector, $\tau_{\ell t}$ is an iceberg trade cost between a manufacturing plant located in ℓ and the U.S., and γ_{jmt} is a location-invariant product–market–year productivity term. The term φ_{ℓ} is a location-specific regulatory-cost shifter defined as the inspection-frequency ratio relative to Home, $\varphi_{\ell} \equiv \nu_{\ell}/\nu_{H}$, where ν_{ℓ} denotes the average inspection frequency in location ℓ (inspections per year) and ν_{H} is the Home benchmark. Hence, $\varphi_{H} = 1$ for the U.S., while locations inspected less frequently have $\varphi_{\ell} < 1$.[54]

Fixed costs. All monetary payoffs in the location-choice model are expressed in millions of dollars. Each manufacturing plant j choosing a production location ℓ incurs a location-specific fixed cost parameter FC_{ℓ}, common to all plants producing in location ℓ. In addition, there is an idiosyncratic plant–market–year shock ϵ^{ℓ}_{jmt} to location profitability. I assume that ϵ^{ℓ}_{jmt} are i.i.d. Type I Extreme Value across locations ℓ and manufacturer–market–year triples (j, m, t), which implies a multinomial logit structure for location choice.[55] Equivalently, supplier j faces the location-specific fixed cost

$$FC^{\ell}_{jmt} = FC_{\ell} - \epsilon^{\ell}_{jmt}. \tag{Eq. FC}$$

The supplier observes its location-profitability shocks before choosing where to produce. I recover the common location-cost components from observed location choices.

4.4. The Manufacturer's Problem

Manufacturers cannot control the realization of a supply shock, but their location and capacity choices determine whether it produces a shortage. Location choice selects a production cost and a capacity-yield distribution; capacity buffers insure contracted demand against low yields. Both decisions are made before shocks occur.

I solve the manufacturer's problem in two steps. Each supplier first chooses a production location. Conditional on locations, suppliers simultaneously choose bid prices and capacity to maximize expected profit. Anticipated profits from the price–capacity problem determine the location choice.

4.4.1. Step 1: Location Choice

Given anticipated Step-2 choices (p^{*}, κ^{*}), manufacturer j makes a discrete choice over the production location ℓ that maximizes expected profits net of fixed costs. The location choice problem is

$$\ell^{*}_{jmt} \in \arg\max_{\ell \in \{H, F_N, F_S\}} \left\{ \mathbb{E}_{\omega \sim F_{\ell}} \left[\pi^{\ell}_{jmt} \left(p^{*}_{jmt}(\ell), \kappa^{*}_{jmt}(\ell); Q^{D}_{jmt}, c^{\ell}_{jmt}(\gamma_{jmt}), \omega \right) \right] - FC_{\ell} + \epsilon^{\ell}_{jmt} \right\}. \tag{Eq. location}$$

[54] More frequent inspections at Home imply higher marginal production costs, all else equal. Inspecting foreign-based plants comes with significant regulatory hurdles; in India and China, FDA inspectors need to announce their visits months in advance as they require the foreign company's sponsorship and the state regulator's approval to obtain visas and cross borders.

[55] This captures plant- and product-specific heterogeneity in location attractiveness (e.g. unobserved contractual relationships, financing conditions, or technological fit).

Each manufacturer trades off three location-specific factors: the capacity-yield distribution F_ℓ, marginal costs c^ℓ_{jmt}, and fixed costs FC_ℓ, together with the idiosyncratic fixed-cost shock ϵ^ℓ_{jmt}. At this stage, the firm observes its productivity draw γ_{jmt}, the location-specific cost shifters entering $c^\ell_{jmt}(\gamma_{jmt})$, and its idiosyncratic fixed-cost shock ϵ^ℓ_{jmt}. The firm knows the location-specific capacity-yield distribution F_ℓ, but does not observe the future realization of the capacity-yield shock ω_{jmt}.

This step delivers an endogenous cost–reliability trade-off: locations with lower marginal costs may have less favorable yield distributions, assigning more probability mass to low surviving-capacity states. For a given demand-to-capacity ratio θ_{jmt}, this raises shortage risk through $F_\ell(\theta_{jmt})$. Firms therefore trade off location-specific cost advantages against expected shortage risk and fixed costs when choosing where to locate.

4.4.2. Step 2: Price and Capacity Choices

Conditional on location ℓ, manufacturer j chooses price and production capacity (p_{jmt}, κ_{jmt}) to maximize expected variable profit, taking as given the location-specific capacity-yield distribution F_ℓ. Using $\theta_{jmt} = Q^D_{jmt}(\delta)/\kappa_{jmt}$, expected profit can be decomposed across non-shortage and shortage states:

$$\mathbb{E}_{\omega\sim F_\ell}\left[\pi^\ell_{jmt}\right] = \underbrace{\left[1 - F_\ell(\theta_{jmt})\right]}_{\text{Pr(no shortage)}} \underbrace{p_{jmt} Q^D_{jmt}(\delta)}_{\text{Revenue | no shortage}} + \underbrace{F_\ell(\theta_{jmt})}_{\text{Pr(shortage)}} \underbrace{p_{jmt}\kappa_{jmt}\mathbb{E}_{\omega\sim F_\ell}\left[\omega \mid \omega < \theta_{jmt}\right]}_{\text{Expected revenue | shortage}} - \underbrace{c^\ell_{jmt}\kappa_{jmt}}_{\text{Variable production cost}} . \tag{12}$$

Equivalently, the Step-2 problem can be written compactly as

$$\max_{p_{jmt},\kappa_{jmt}} \mathbb{E}_{\omega\sim F_\ell}\left[\pi^\ell_{jmt}\right] = \max_{p_{jmt},\kappa_{jmt}} \left\{ p_{jmt}\left(\left[1 - F_\ell(\theta_{jmt})\right] Q^D_{jmt}(\delta) + \kappa_{jmt} h_\ell(\theta_{jmt})\right) - c^\ell_{jmt}\kappa_{jmt} \right\}, \quad \text{(Eq. Profit)}$$

where $h_\ell(\theta_{jmt})$ is the probability-weighted surviving-capacity share in shortage states

$$h_\ell(\theta_{jmt}) \equiv \int_0^{\theta_{jmt}} \omega f_\ell(\omega)\, d\omega = F_\ell(\theta_{jmt})\mathbb{E}_{\omega\sim F_\ell}\left[\omega_{jmt} \mid \omega_{jmt} < \theta_{jmt}\right], \tag{13}$$

and f_ℓ denotes the pdf associated with F_ℓ. This function is weakly increasing, $h'_\ell(\theta) = \theta f_\ell(\theta) \geq 0$, and since $\theta \in (0, 1]$, bounded by $0 \leq h_\ell(\theta) \leq \theta F_\ell(\theta) \leq F_\ell(\theta)$.

Manufacturer j's expected revenue is the contracted price times expected realized market supply. In the no-shortage state, realized market supply equals the contracted quantity $Q^D_{jmt}(\delta)$. In the shortage state, realized market supply equals surviving capacity $\kappa_{jmt}\omega_{jmt}$. Expected profit is therefore an average of profits in shortage and no-shortage states, weighted by the probability of each state.

Manufacturers invest in production capacity $\kappa_{jmt} \geq 0$ before shocks are realized, which comes at a cost of c^ℓ_{jmt} per unit.[56] Capacity adjustments are lumpy; firms commit capacity κ_{jmt} for each market,

[56] I assume no extra disposal fee: manufacturers "freely" dispose of any unsold capacity at the end of the contract period. This assumption helps with tractability, while allowing capture of the key feature of interest: producing surplus is costly in these low-profit-margin markets, and manufacturers may want to minimize losses by reducing buffers.

which cannot be increased within the period.[57] Manufacturers pay this marginal cost per unit of ex-ante capacity, regardless of the realized yield. Manufacturers can insure against shocks by building more capacity than the contracted (demanded) quantities, thus building costly capacity buffers:

$$B_{jmt} \equiv \kappa_{jmt} - Q^D_{jmt}(\delta). \tag{Eq. Buffers}$$

4.5. Equilibrium Conditions

Solving the manufacturer's problem yields two first-order conditions that jointly determine equilibrium prices and capacity. Appendix A.2 provides the full derivation of these conditions. Recall that $\theta_{jmt} = Q^D_{jmt}(\delta)/\kappa_{jmt}$ denotes the demand-to-capacity ratio and that $h_\ell(\theta_{jmt}) = \int_0^{\theta_{jmt}} \omega \, dF_\ell(\omega)$ is the probability-weighted surviving-capacity share in shortage states, both defined in Eq. (6) and Eq. (13).

Capacity FOC. The first-order condition with respect to capacity is

$$p^*_{jmt} = \frac{c^\ell_{jmt}}{h_\ell(\theta^*_{jmt})}. \tag{FOC 1}$$

Price FOC. Under my demand specification, the own-price elasticity of latent demand is

$$\eta_{jjmt} \equiv \frac{\partial Q^D_{jmt}(\delta)}{\partial p_{jmt}} \frac{p_{jmt}}{Q^D_{jmt}(\delta)} = -\alpha(1 - s^D_{jmt}).^{58}$$

Using this derivative, the first-order condition with respect to price can be written as

$$\kappa^*_{jmt} = -\left(1 + \eta^*_{jjmt}\right) \frac{1 - F_\ell(\theta^*_{jmt})}{h_\ell(\theta^*_{jmt})} Q^D_{jmt}(\delta^*), \tag{FOC 2}$$

or, dividing by $Q^D_{jmt}(\delta^*)$,

$$\frac{1}{\theta^*_{jmt}} = -\left(1 + \eta^*_{jjmt}\right) \frac{1 - F_\ell(\theta^*_{jmt})}{h_\ell(\theta^*_{jmt})}. \tag{FOC 2'}$$

The capacity condition equates the unit cost of capacity to its expected marginal revenue: additional capacity increases deliveries only in shortage states. The price condition balances the revenue from a higher bid against the loss of contracted demand, accounting for whether capacity is binding. Together, the conditions determine price and the buffer chosen before shocks occur.

[57] The generic sterile injectable industry is well-known for practicing just-in-time production, so that I treat production lines as operating continuously at their maximal capacity. Manufacturers choose how many lines to allocate to drug product m at the beginning of the period and then run their lines to maximal capacity. In reality, manufacturers allegedly do not keep more than one to two weeks of stocks. Since the Covid-19 pandemic, the FDA has acted to support legislation that would force drug manufacturers to build at least three months of stocks, a buffer that many health policymakers still judge insufficient given the average duration of shortage events. As of October 2023, this legislation has not been enacted.

[58] The corresponding derivative of latent demand is $\frac{\partial Q^D_{jmt}(\delta)}{\partial p_{jmt}} = -\frac{\alpha}{p_{jmt}} Q^D_{jmt}(\delta)(1 - s^D_{jmt})$.

Markup form. Combining the two first-order conditions gives

$$p^*_{jmt} = \underbrace{-\left(\frac{1}{1+\eta^*_{jjmt}}\right) c^{\ell}_{jmt}}_{\text{cost–elasticity term}} \underbrace{\left(\frac{1}{\theta^*_{jmt}\left[1-F_{\ell}(\theta^*_{jmt})\right]}\right)}_{\text{supply-risk adjustment}}. \tag{14}$$

4.6. Consumer Welfare

I measure consumer welfare using the inclusive value implied by the logit demand system and adjust it for the ex-post supply shortfalls generated by shocks. In the absence of shortages, this object is the standard logit consumer surplus index (Anderson, de Palma, and Thisse, 1992; Capps, Dranove, and Satterthwaite, 2003). Supply-driven shortages create a wedge between ex-ante welfare under the logit choice set and realized, post-shock welfare: some buyers who initially choose an inside good are forced into the outside option when realized supply falls short of latent demand.

4.6.1. Consumer Surplus without Shortages

Consider market m in period t, and let δ_{mt} denote the vector of supplier mean utilities defined in Eq. (2). With i.i.d. Type-I extreme value taste shocks, the expected maximum utility of a potential buyer is

$$EU_{mt}(\delta) = \mathbb{E}\left[\max\{\varepsilon_{i0mt}, \delta_{1mt}+\varepsilon_{i1mt}, \ldots, \delta_{J_{mt}mt}+\varepsilon_{iJ_{mt}mt}\}\right] = \gamma_E + \ln\left(1+\sum_{j\in J_{mt}} \exp\left(\delta_{jmt}\right)\right), \tag{15}$$

where $\gamma_E \approx 0.577$ is the Euler–Mascheroni constant (mean of the logit error term).[59] The Euler constant cancels in counterfactual differences, so the relevant frictionless consumer surplus is the inclusive value

$$\boxed{CS^{\text{logit}}_{mt}(\delta) \equiv \ln\left(1+\sum_{j\in J_{mt}} \exp\left(\delta_{jmt}\right)\right) = -\ln s^D_{0mt}(\delta)} \tag{16}$$

where

$$s^D_{0mt}(\delta) = \frac{1}{1+\sum_{j\in J_{mt}} \exp\left(\delta_{jmt}\right)}$$

is the latent outside-option share implied by demand before supply shocks realize. Equation (16) shows that consumer surplus falls when the latent outside share rises. The superscript D emphasizes that this outside share is based on latent demand, not realized market supply. Because the demand specification uses log prices, consumer surplus is reported in expected-utility units rather than dollars. It should therefore be interpreted as a welfare index, not as a money-metric compensating variation.[60]

[59] In Eq. (2), the outside option has mean utility normalized to zero, $\delta_{0mt} = 0$, so $v_{i0mt} = \varepsilon_{i0mt}$.

[60] Unlike in a linear-price specification, the coefficient $\alpha > 0$ is not a constant marginal utility of income. For this reason, I do not divide logit inclusive values by α when constructing welfare. A money-metric version requires a separate price-equivalent conversion rather than division by α.

4.6.2. Consumer Surplus with Shortages (Excess Demand)

In the model, consumers choose their drug products (and thus $Q^D(\delta)$) *ex-ante* expecting full availability. Supply shocks are realized *ex post*. Consumers who face a shortage for their chosen drug are forced into the outside option (no or delayed treatment) without the opportunity to re-optimize their initial choice probabilities.[61] *Realized* welfare is therefore the weighted sum of utilities for unaffected and shortage-displaced consumers, and it is weakly below the ex-ante logit welfare.

Realized and Excess Outside Shares. For market-time (m,t), define the *realized fill rate* for drug manufacturer j as the ratio of realized market supply to latent demand:

$$r_{jmt} \equiv \frac{q^{\text{real}}_{jmt}}{Q^D_{jmt}(\delta)} = \min\left\{1, \frac{\kappa_{jmt}\omega_{jmt}}{Q^D_{jmt}(\delta)}\right\} = 1 - s^{\text{excess}}_{jmt}(\omega) \in [0,1]. \tag{17}$$

If $r_{jmt} = 1$, demand is fully met; if $r_{jmt} < 1$, a fraction $1 - r_{jmt}$ of intended demand is shortage-displaced. The realized outside share s^{real}_{0mt} is the total share of buyers who end up in the outside option after supply shocks realize:

$$s^{\text{real}}_{0mt} = 1 - \frac{1}{M_{mt}} \sum_{j \in J_{mt}} q^{\text{real}}_{jmt} = 1 - \sum_{j \in J_{mt}} r_{jmt} s^D_{jmt}(\delta). \tag{18}$$

It is weakly larger than the latent outside share $s^D_{0mt}(\delta)$, with the *excess outside share* given by

$$s^{\text{real}}_{0mt} - s^D_{0mt}(\delta) = \sum_{j \in J_{mt}} s^D_{jmt}(\delta)(1 - r_{jmt}) = \left[1 - s^D_{0mt}(\delta)\right] s^{\text{excess}}_{mt}(\omega). \tag{19}$$

Thus s^{excess}_{mt} is the market-level share of excess demand defined in (Eq. Market Excess), whereas $s^{\text{real}}_{0mt} - s^D_{0mt}$ is the share of the full potential market forced into the outside option by shortages.

Utility of Unaffected and Shortage-Displaced Buyers. Consumers who receive their chosen inside good, as well as consumers who originally chose the outside option, obtain the standard *ex ante* expected utility

$$E[U_{\text{unaffected}}] = E[U_{ij} \mid \text{chose } j] = \gamma_E + CS^{\text{logit}}_{mt}(\delta) = \gamma_E - \ln s^D_{0mt}(\delta). \tag{20}$$

Shortage-displaced buyers are those who initially choose an inside good but are forced into the outside option after supply shocks realize. These buyers receive the outside-option utility conditional on having rejected the outside option ex ante. As shown in Appendix A.3, this conditional utility is

$$E[U_{\text{displaced}}] = \gamma_E - \frac{s^D_{0mt}(\delta)}{1 - s^D_{0mt}(\delta)} CS^{\text{logit}}_{mt}(\delta). \tag{21}$$

This utility is below the unconditional mean outside-option utility, γ_E, because shortage-displaced buyers are forced into an alternative they had rejected ex ante.

[61] The model treats shortages as ex-post supply shortfalls under fixed procurement contracts. If a hospital contracts with supplier j and that supplier is disrupted, the hospital cannot costlessly reallocate the same demand to another supplier within the contracting period.

Utility Cost of Shortages. The loss per shortage-displaced buyer is the difference between the expected utility of receiving the initially chosen inside good and the expected utility of being forced into the outside option. I denote this average shortage loss by

$$\bar{\mathcal{L}}_{mt}(\delta) \equiv E[U_{\text{unaffected}}] - E[U_{\text{displaced}}] = \frac{-\ln s^D_{0mt}(\delta)}{1 - s^D_{0mt}(\delta)} = \frac{CS^{\text{logit}}_{mt}(\delta)}{1 - s^D_{0mt}(\delta)}. \tag{22}$$

This object is the average welfare loss, in expected-utility units, for each buyer displaced into the outside option by a shortage.

Realized Consumer Surplus with Ex-Post Shortages. Using the expressions above for unaffected and shortage-displaced buyers, "realized" consumer surplus in market (m,t) equals the frictionless logit surplus minus the welfare loss from buyers forced into the outside option:

$$\boxed{CS^{\text{real}}_{mt}(\delta, s^{\text{real}}_{0mt}) \equiv \underbrace{CS^{\text{logit}}_{mt}(\delta)}_{\text{frictionless logit surplus}} - \underbrace{\left[s^{\text{real}}_{0mt} - s^D_{0mt}(\delta)\right] \bar{\mathcal{L}}_{mt}(\delta)}_{\text{shortage penalty}} = CS^{\text{logit}}_{mt}(\delta) \left[\frac{1 - s^{\text{real}}_{0mt}}{1 - s^D_{0mt}(\delta)}\right]} \tag{23}$$

The term $s^{\text{real}}_{0mt} - s^D_{0mt}(\delta)$ is the share of buyers who would have purchased an inside good absent supply shocks but are forced into the outside option. Thus, realized consumer surplus is the standard logit surplus index, measured in expected-utility units, net of the mass of shortage-displaced buyers valued at the average shortage loss $\bar{\mathcal{L}}_{mt}(\delta)$.

Welfare Loss with Ex-Post Shortages. The total welfare loss due to shortages is the difference between realized consumer surplus and the frictionless logit surplus:

$$WL^{\text{short}}_{mt} \equiv CS^{\text{logit}}_{mt}(\delta) - CS^{\text{real}}_{mt}(\delta, s^{\text{real}}_{0mt}) = \left[s^{\text{real}}_{0mt} - s^D_{0mt}(\delta)\right] \bar{\mathcal{L}}_{mt}(\delta) \geq 0. \tag{24}$$

Equivalently, using supplier-level fill rates r_{jmt},

$$WL^{\text{short}}_{mt} = \left[\sum_{j=1}^{J_{mt}} s^D_{jmt}(\delta)(1 - r_{jmt})\right] \bar{\mathcal{L}}_{mt}(\delta). \tag{25}$$

This expression makes clear that shortage losses are larger when disruptions affect high-share suppliers, when realized fill rates are low, or when the outside option is especially unattractive. Section 5 recovers the demand, capacity-yield, cost, and location-choice primitives needed to quantify these mechanisms.

5. Estimation of the Structural Model

I estimate the model in four steps. First, I recover demand while treating sales during shortages as censored. Second, conditional on demand, I estimate location-specific capacity-yield distributions using shortage incidence, the mean annual shortage share, and its variance conditional on a shortage. Third, I recover marginal costs from the capacity first-order condition and estimate trade-cost and productivity parameters. Finally, I estimate location fixed-cost indices from observed production choices and

candidate-location expected profits. Table 1 summarizes the procedure; Appendix Table B.3 reports the estimates.

Table 1: **Structural Parameters and Estimation Methods**

Step	Parameter Description	Symbol	Estimation Method
Step 1: Logit Demand under Censoring			
	Price sensitivity	α	IV-2SLS (Within-Estimator)
	Brand-status and presentation controls	$\beta_{\text{brand}}, \beta_{\text{bg}}, \beta_A$	IV-2SLS (Within-Estimator)
	Demand-shock distribution and market size	$\mu_\xi, \sigma_\xi, M_{mt}$	EM-style Censored MLE
Step 2: Supply Shocks			
	Capacity-yield shock distribution (Beta)	$\{a_{\omega\ell}, b_{\omega\ell}\}_\ell$	Two-step GMM matching shortage moments
Step 3: Trade Costs and TFP			
	Trade–cost parameters	$\phi^{\text{dist}}, \phi^{\text{dist}\times\text{oil}}$	OLS on recovered marginal costs
	TFP distribution parameters	$(\mu_\gamma, \sigma_\gamma)$	Method of Moments
Step 4: Location Choice			
	Fixed-cost parameters	$\{FC_{H,\mathcal{T}}, FC_{F_S,\mathcal{T}}\}_{\mathcal{T}=1}^{4}$	Conditional-logit MLE

Notes: $\ell \in \{H, F_N, F_S\}$ indexes Home (U.S.), Foreign North (OECD and other high-income foreign locations), and Foreign South (Asian locations, primarily India and China). In Step 4, $\mathcal{T}$ indexes the four fixed-cost blocks 2005–2010, 2011–2013, 2014–2016, and 2017–2019. The fixed-cost normalization is $FC_{F_N,\mathcal{T}} = 0$ in every block.

5.1. Sample Definition

The estimation sample covers 2005–2019. I define a market m as an ATC5 chemical-substance class within the injectable sample, observed in calendar year t. Products j are defined at the National Drug Code (NDC) 5-4 level, aggregating commercial package sizes.[62] Each product is linked to its manufacturing plant and to one of three production regions ℓ: Home (H), Foreign North (F_N), and Foreign South (F_S). Home denotes U.S. plants; Foreign North denotes OECD and other high-income foreign locations; and Foreign South denotes Asian locations, primarily India and China.

For each drug product j, I construct prices and realized quantities using IQVIA sales data. IQVIA reports sales revenue and physical units at the package-level NDC (NDC 5-4-2). For injectables, I use "eaches" as the unit of quantity, corresponding to individual vials, ampules, or syringes. I aggregate package presentations to the NDC 5-4–year level by summing sales revenue and eaches across all package-level NDCs associated with product j. I then define the unit price as

$$p_{jmt} = \frac{\sum_{h \in \mathcal{P}(j)} \text{Sales}_{hjmt}}{\sum_{h \in \mathcal{P}(j)} \text{Eaches}_{hjmt}},$$

where $\mathcal{P}(j)$ denotes the set of package-level NDCs associated with product j. Realized quantities are defined as total IQVIA eaches sold in the U.S. during year t.[63] Additional details on drug market

[62] This definition aggregates separate commercial package sizes corresponding to the same labeler-product-dosage-form. Each drug can be offered by a manufacturer in multiple presentations; I treat these as one product and include the number of presentations as a product characteristic, as in Dubois and Lasio (2018) and Atal, Cuesta, and Sæthre (2026).

[63] IQVIA sales data do not reveal hospital-specific procurement contracts or rebates. I therefore interpret p_{jmt} as an average realized unit price at the product-year level. In low-margin generic injectable markets, unobserved rebates are likely to be small relative to branded-drug markets.

characteristics, and the distribution of price, shortages, and sales across markets and locations are reported in Appendix B.

5.2. Step 1: Demand Estimation

Shortages create a censoring problem for demand estimation. Standard discrete-choice methods equate observed sales with demanded quantities; in this setting, that equality holds only for unconstrained (non-short) suppliers. For a supplier in shortage, observed sales are constrained by realized supply:

$$q_{jmt}^{\text{real}} = \min\left\{Q_{jmt}^{D}(\delta),\, \kappa_{jmt}\omega_{jmt}\right\} \leq Q_{jmt}^{D}(\delta), \tag{26}$$

where $Q_{jmt}^{D}(\delta)$ is latent demand, κ_{jmt} is capacity, and ω_{jmt} is the location-specific capacity-yield shock. Inverting observed shares as if they measured demand would attribute shorted quantities to low demand: it would understate latent demand for products in shortage, contaminate the recovered demand shocks ξ_{jmt}, and bias the substitution patterns that the supply model takes as inputs.[64]

To address this, I build a two-stage demand estimation strategy. Stage 1 identifies the slope parameters of utility from within-market variation. Stage 2 recovers the potential market size M_{mt} and the distribution of demand shocks from a censored likelihood, and imputes latent demand for constrained (short) products via an iterative EM algorithm. Because shortage status is an equilibrium outcome, the two stages are iterated to convergence.

Stage 1: Slope parameters. Taking logs in equation (4) and differencing against the outside option, the demand system of Section 4.2 can be written in quantities as

$$\ln Q_{jmt}^{D}(\delta) = \mu_{mt} - \alpha \ln p_{jmt} + \boldsymbol{\beta}'\mathbf{X}_{jmt} + \xi_{jmt}, \qquad \mu_{mt} \equiv \ln Q_{0mt}^{D}(\delta), \tag{27}$$

where $\mathbf{X}_{jmt}$ contains the brand-status indicators, with generics omitted, and product-line breadth. The market–year intercept $\mu_{mt} = \ln Q_{0mt}^{D}$ is the log of latent outside-option demand. I absorb it with ATC5-by-year fixed effects and identify $(\alpha, \boldsymbol{\beta})$ from within-market variation across products. This avoids specifying market size when estimating the slopes; market size and latent demand are recovered jointly in Stage 2.[65] The fixed effects also absorb common market–year demand shifters.

For unconstrained observations, $q_{jmt}^{\text{real}} = Q_{jmt}^{D}(\delta)$, so equation (27) can be estimated by two-stage least squares on the subsample of product–years not in shortage. Selection into that subsample depends on the demand shock itself: given capacity, high-ξ products are more likely to be rationed. The procedure therefore iterates with Stage 2. After the initial pass on unconstrained observations, constrained observations re-enter the regression at their imputed latent quantities, and the reported estimates are those of the converged pass.

[64] Conlon and Mortimer (2013) address a related problem when availability is observed at high frequency and consumers re-optimize over the realized choice set. Here rationing is partial within the contracting period and buyers cannot re-optimize (Section 4.6), so I instead treat constrained sales as censored observations of latent demand.

[65] In the fixed-coefficient logit, latent outside-option demand enters the log-quantity equation as a common market–year intercept, which drops out of within-market comparisons (Huang and Rojas, 2014; Dubois and Lasio, 2018).

Price endogeneity. Prices may be correlated with unobserved demand ξ_{jmt}. I instrument log prices with differentiation instruments in the spirit of Gandhi and Houde (2019): counts and distance-weighted counts of rival products, distinguishing rivals in the same ATC5 market from those in other ATC5 markets within the same ATC4 class. Rival sets are measured four quarters earlier and exclude products sold by the same labeler. Identification requires these instruments to be orthogonal to current product-level demand shocks conditional on market–year fixed effects and observed characteristics. Appendix B.5 describes their construction.

Stage 2: Market size and censored demand shocks. Conditional on candidate market sizes and the slope estimates, equation (27) recovers ξ_{jmt} for unconstrained products and a lower bound $\underline{\xi}_{jmt}$ for constrained products. I assume $\xi_{jmt} \sim \mathcal{N}(\mu_\xi, \sigma_\xi^2)$ and estimate its distribution by censored maximum likelihood. An EM algorithm alternates between three operations: recovering the market size M_{mt} from a residual-root calibration, under which model-implied demand for unconstrained products aggregates exactly to their observed sales; updating (μ_ξ, σ_ξ) from the censored likelihood; and replacing constrained sales with expected latent demand under the truncated shock distribution. Upon convergence, the algorithm delivers latent quantities $\hat{Q}^D_{jmt}$, market sizes $\hat{M}_{mt}$, latent shares s^D_{jmt}, and the latent and observed outside shares $(s^D_{0mt}, s^{\text{real}}_{0mt})$ that enter the consumer-surplus expressions in Section 4.6. Appendix B.4 reports the algorithm.

Outputs. With log prices, the own- and cross-price elasticities of latent demand are

$$\eta_{jjmt} = -\hat{\alpha}\big(1 - s^D_{jmt}\big), \qquad \eta_{jkmt} = \hat{\alpha}\, s^D_{kmt},$$

computed at the latent shares, so that substitution patterns are not confounded by supply constraints. The latent quantities $\hat{\mathbf{Q}}^D$, market sizes $\hat{M}_{mt}$, and the wedge between latent and observed outside shares are the demand-side inputs to the supply estimation in Section 5.3 and to the welfare calculations in Section 4.6. Parameter estimates, first-stage diagnostics, and implied elasticities are reported in Appendix B.3.

5.3. Supply Estimation

Conditional on recovered latent demand (Step 1), I estimate the remaining structural objects in three steps. Step 2 estimates the location-specific capacity-yield distributions from shortage moments. Step 3 recovers marginal costs from the capacity first-order condition and uses those marginal costs to estimate trade costs and productivity. Step 4 estimates fixed-cost wedges in the location-choice problem using a conditional-logit likelihood.

5.3.1. Step 2: Location-Specific Capacity-Yield Shocks

Conditional on the demand estimates, the recovered latent demand $\widehat{Q}^D_{jmt}$, and the estimated own-price elasticity $\widehat{\eta}_{jjmt}$, I estimate the distribution of supplier-level capacity-yield shocks in each production

location. Let $\ell(j)$ denote product j's observed plant location, which is fixed in estimation; the structural location choice ℓ_{jmt} may change in the counterfactuals. I parameterize the location-specific yield distribution F_ℓ as

$$\omega_{jmt} \mid \ell(j) = \ell \sim \text{Beta}(a_{\omega\ell}, b_{\omega\ell}), \qquad \ell \in \{H, F_N, F_S\}, \tag{28}$$

The parameter vector is

$$\Theta_\omega \equiv \{(a_{\omega\ell}, b_{\omega\ell}) : \ell \in \{H, F_N, F_S\}\}.$$

Moment Conditions. I match three moments of the annual shortage distribution within each location. Let $\rho^{\text{obs}}_{jmt} \in \{0,1\}$ indicate whether product j experiences any shortage during year t, and let $s^{\text{excess,obs}}_{jmt} \in [0,1]$ denote the observed annual shortage share, measured as the fraction of the year during which the product is in shortage. Under the uniform-flow assumption introduced in the model, $s^{\text{excess,obs}}_{jmt}$ is the data counterpart of the model's share of excess demand $s^{\text{excess}}_{jmt}(\omega)$ defined in Eq. (10), and $\rho^{\text{obs}}_{jmt} \in \{0,1\}$ is the counterpart of $\mathbf{1}\{s^{\text{excess}}_{jmt}(\omega) > 0\}$.

For each location ℓ, I target three moment restrictions:

$$\mathbb{E}\left[\rho^{\text{obs}}_{jmt} \mid \ell(j) = \ell\right] = \mathbb{E}_{jmt}\left[F_\ell(\theta_{jmt}) \mid \ell(j) = \ell\right], \tag{Shock Moment 1}$$

$$\mathbb{E}\left[s^{\text{excess,obs}}_{jmt} \mid \ell(j) = \ell\right] = \mathbb{E}_{jmt}\left[F_\ell(\theta_{jmt}) - \frac{h_\ell(\theta_{jmt})}{\theta_{jmt}} \,\middle|\, \ell(j) = \ell\right], \tag{Shock Moment 2}$$

$$\text{Var}\left(s^{\text{excess,obs}}_{jmt} \,\middle|\, \rho^{\text{obs}}_{jmt} = 1, \ell(j) = \ell\right) = \text{Var}_{jmt,\omega}\left(s^{\text{excess}}_{jmt}(\omega) \,\middle|\, s^{\text{excess}}_{jmt}(\omega) > 0, \ell(j) = \ell\right). \tag{Shock Moment 3}$$

On the model side, θ_{jmt} is the equilibrium demand-to-capacity ratio implied by the candidate Θ_ω. Expectations and variances integrate over the candidate yield distribution and the empirical distribution of product–market–year observations within location. The first two moments discipline the probability and average severity of shortages; the third disciplines the dispersion of shortage severity among products that do go short. Appendix B.6.1 derives the mapping between shortage duration and first and second moments, and Appendix B.7 provides the exact sample analogs.

Stacking the three moments across the three locations yields nine moment conditions for the six parameters in Θ_ω. I estimate Θ_ω by two-step GMM:

$$\widehat{\Theta}_\omega = \arg\min_{\Theta_\omega} g_N(\Theta_\omega)' W_N g_N(\Theta_\omega), \tag{29}$$

where $g_N(\Theta_\omega)$ stacks the location-specific sample moment differences and W_N is the second-step weighting matrix.

Nested Fixed-Point Algorithm. The model-implied moments depend on the equilibrium demand-to-capacity ratio θ_{jmt}, which is endogenous because the firm's capacity choice responds to the yield distribution. The entire algorithm is detailed in Appendix B.7. For each candidate Θ_ω and each observation in the supply-estimation sample, I solve the equilibrium condition

$$\frac{1}{\theta_{jmt}} = -\left(1 + \widehat{\eta}_{jjmt}\right) \frac{1 - F_\ell(\theta_{jmt})}{h_\ell(\theta_{jmt})}, \qquad \ell = \ell(j). \tag{30}$$

I then recover the capacity implied by the candidate distribution as $\kappa_{jmt}(\Theta_\omega) = \widehat{Q}^D_{jmt}/\theta_{jmt}(\Theta_\omega)$, evaluate the model-implied moments, and update Θ_ω until the GMM criterion converges. Solving the inner problem directly in θ_{jmt} is convenient because it is the sufficient statistic entering all three moments. Upon convergence, I recover planned capacity and the capacity buffer as

$$\widehat{\kappa}_{jmt} = \frac{\widehat{Q}^D_{jmt}}{\theta_{jmt}(\widehat{\Theta}_\omega)}, \qquad \widehat{B}_{jmt} = \widehat{\kappa}_{jmt} - \widehat{Q}^D_{jmt}. \tag{31}$$

5.3.2. Step 3: Recovering Marginal Costs, Trade Costs, and Productivity

In Step 3, I recover firms' marginal costs, trade costs and firms' productivity shocks (TFP) sequentially.

Marginal Costs. Conditional on the estimated demand and capacity-yield distributions from the previous steps, I recover the model-implied marginal cost for product j in market m and year t by inverting the capacity first-order condition (FOC 1):

$$\widehat{c}^{\ell}_{jmt} = p_{jmt}\, h_\ell\left(\widehat{\theta}_{jmt}\right) = p_{jmt} \int_0^{\widehat{\theta}_{jmt}} \omega f_\ell(\omega)\, d\omega, \tag{32}$$

where p_{jmt} is the IQVIA realized unit price defined above and $\widehat{\theta}_{jmt}$ is the demand-to-capacity ratio recovered in Step 2. This inversion uses the structural relationship between prices, capacities, and the location-specific capacity-yield distribution.

Trade Costs. I parameterize iceberg trade costs as a function of geographic distance and fuel costs:[66]

$$\ln \tau_{\ell(j)t} = \phi_0 + \phi^{\text{dist}} \ln \text{dist}_{\ell(j)} + \phi^{\text{dist}\times\text{oil}} \left(\ln \text{dist}_{\ell(j)} \cdot \ln \text{oil}_t\right), \tag{33}$$

where $\ell(j)$ denotes the location of the manufacturing plant linked to product j, $\text{dist}_{\ell(j)}$ is the Haversine distance from that plant to the U.S. market, and oil_t is a global bunker-fuel price index in year t. This specification allows the relationship between distance and trade costs to vary with fuel prices.

Because marginal costs identify trade costs and productivity only through their ratio, I normalize the common log trade-cost level to zero, $\phi_0 = 0$. The corresponding common cost level is absorbed into productivity. This normalization does not affect the estimated distance coefficients or counterfactual marginal-cost comparisons.

Substituting (33) into the structural marginal-cost equation (Eq. MC) gives the estimating equation

$$\ln \widehat{c}^{\ell}_{jmt} - \ln w_{\ell t} - \ln \varphi_\ell = \phi^{\text{dist}} \ln \text{dist}_{\ell(j)} + \phi^{\text{dist}\times\text{oil}} \left(\ln \text{dist}_{\ell(j)} \cdot \ln \text{oil}_t\right) + \chi_m + \chi_t + u_{jmt}, \tag{34}$$

where χ_m are molecule–form fixed effects (controlling for the complexity of the drug), and χ_t are year fixed effects (capturing global input-cost shocks not explained by wages or oil). The fixed effects also absorb the common regression level under the usual fixed-effect normalization. The term u_{jmt} is the remaining location-invariant cost component. Thus, the fixed effects and residual jointly parameterize $-\ln \gamma_{jmt}$; the regression residual alone should not be interpreted as productivity.

[66] U.S. import tariffs on finished pharmaceutical products (HS 3004) are essentially zero and unchanged over the 2005–2019 sample period, so I do not include tariffs in the trade-cost function.

I estimate (34) by OLS, treating $\widehat{c}^{\ell}_{jmt}$ as a generated dependent variable.[67]

The distance coefficient is identified from cost differences across plants at different distances from the U.S., conditional on the fixed effects and observed cost shifters. The distance–oil interaction is identified from changes in this relationship as oil prices vary. Identification requires the residual cost component to be mean-independent of distance and its interaction with oil prices, conditional on the included controls.

Total Factor Productivity. Given the structural marginal-cost equation, the recovered marginal cost $\widehat{c}^{\ell}_{jmt}$, and the predicted trade cost $\widehat{\tau}_{\ell(j)t}$, I recover productivity at the product–market–year level as

$$\widehat{\gamma}_{jmt} = \frac{w_{\ell t}\,\widehat{\tau}_{\ell(j)t}\,\varphi_{\ell}}{\widehat{c}^{\ell}_{jmt}}. \tag{35}$$

$$\text{Equivalently,} \qquad \ln \widehat{\gamma}_{jmt} = \ln \widehat{\tau}_{\ell(j)t} - \left(\ln \widehat{c}^{\ell}_{jmt} - \ln w_{\ell t} - \ln \varphi_{\ell}\right).$$

Here ℓ is the observed plant location associated with product j, $w_{\ell t}$ is the location-year pharmaceutical-manufacturing wage, and φ_{ℓ} is the regulatory-cost shifter calibrated from relative inspection frequencies.[68] Because γ_{jmt} is defined at the structural product–market–year level, I retain one productivity observation per (j, m, t) and do not aggregate productivity to the physical plant–year level.

Under the trade-cost normalization above, I assume that productivity follows one common log-normal distribution across locations:

$$\ln \gamma_{jmt} \sim \mathcal{N}\left(\mu_{\gamma}, \sigma^2_{\gamma}\right). \tag{36}$$

I estimate $(\mu_{\gamma}, \sigma_{\gamma})$ by matching the empirical mean and variance of $\ln \widehat{\gamma}_{jmt}$ across the estimation sample. In the location-choice estimation and counterfactuals, I hold each recovered productivity realization $\widehat{\gamma}_{jmt}$ fixed across candidate locations.

5.3.3. Step 4: Location Fixed Costs

The final step estimates the common component of location fixed costs from observed production-location choices. The estimator is a conditional-logit model at the product–market–year level.

For each product–market–year observation (j, m, t) and candidate location ℓ, I construct expected variable profit $\mathbb{E}_{\ell}[\pi^{\ell}_{jmt}]$, where $\mathbb{E}_{\ell}[\cdot] \equiv \mathbb{E}_{\omega \sim F_{\ell}}[\cdot]$ denotes the expectation over the capacity-yield shock, with the recovered productivity $\widehat{\gamma}_{jmt}$ held fixed. For each candidate location, I solve product j's price and capacity problem under that location's wages, trade costs, regulatory costs, and capacity-yield distribution, taking demand conditions and competitors' choices as given and holding product productivity fixed across candidate locations. Differences in $\mathbb{E}_{\ell}[\pi^{\ell}_{jmt}]$ therefore compare the profitability of producing the same product for the same market under alternative regional operating conditions.

[67] Standard errors account for the generated supply-side objects using the bootstrap procedure described in Appendix Section B.7.2.

[68] Average FDA inspection frequencies are approximately one inspection every 2.5 years for Home, every 5 years for Foreign North, and every 14 years for Foreign South. Equivalently, $\nu_H = 1/2.5$, $\nu_{F_N} = 1/5$, and $\nu_{F_S} = 1/14$ inspections per year. Because $\varphi_{\ell} \equiv \nu_{\ell}/\nu_H$, this gives $\varphi_H = 1$, $\varphi_{F_N} \simeq 0.50$, and $\varphi_{F_S} \simeq 0.18$.

The payoff index from locating production in region ℓ is

$$V_{jmt}^{\ell} = \beta_{\pi}\, \mathbb{E}_{\ell}[\pi_{jmt}^{\ell}] - FC_{\ell,\mathcal{T}(t)} + \epsilon_{jmt}^{\ell}, \qquad \epsilon_{jmt}^{\ell} \sim \text{EV1}(0,1). \tag{37}$$

where ϵ_{jmt}^{ℓ} is i.i.d. Type-I extreme value with unit scale and $\beta_{\pi} > 0$ is the inverse scale of the monetary location-profitability shock in equation (Eq. FC). Profits are measured in millions of dollars. Multiplying the payoffs in equation (Eq. location) by β_{π} gives this index. In this section, $FC_{\ell,\mathcal{T}(t)}$ denotes the normalized fixed cost: the tables report indices, and $FC_{\ell,\mathcal{T}(t)}/\beta_{\pi}$ is the corresponding monetary location-cost wedge in millions of dollars. The function $\mathcal{T}(t)$ assigns year t to one of four periods: {2005–2010, 2011–2013, 2014–2016, 2017–2019}. The implied probability that production is located in region ℓ is

$$P_{jmt}^{\ell} = \frac{\exp\left[\beta_{\pi}\, \mathbb{E}_{\ell}[\pi_{jmt}^{\ell}] - FC_{\ell,\mathcal{T}(t)}\right]}{\sum_{r \in \{H, F_N, F_S\}} \exp\left[\beta_{\pi}\, \mathbb{E}_{r}[\pi_{jmt}^{r}] - FC_{r,\mathcal{T}(t)}\right]}. \tag{38}$$

Only differences in the location-cost indices across regions are identified. I therefore normalize Foreign North to zero in every period:[69]

$$FC_{F_N,\mathcal{T}} = 0, \qquad \mathcal{T} = 1, \ldots, 4.$$

Let ℓ_{jmt}^{obs} denote the observed production location. Conditional on β_{π}, I estimate the eight remaining location–period indices by maximum likelihood:

$$\widehat{FC}(\beta_{\pi}) = \arg \max_{\{FC_{H,\mathcal{T}}, FC_{F_S,\mathcal{T}}\}_{\mathcal{T}=1}^{4}} \sum_{j,m,t} \log P_{jmt}^{\ell_{jmt}^{\text{obs}}}. \tag{39}$$

The coefficient β_{π} governs the responsiveness of production locations to expected-profit differences. I calibrate its benchmark value using the dispersion of pairwise candidate-profit differences. Appendix B.7.1 describes the calibration and inference and reports sensitivity to weaker location responses.

5.4. Estimation Results

Table B.3 reports the full demand and supply estimates, and Table B.7 reports the main model moments.

Demand estimates. The estimated price-sensitivity parameter is $\widehat{\alpha} = 1.264$, and the implied mean own-price elasticity of latent demand is -1.17. Brand status and product-line breadth have positive estimated coefficients (Appendix Table B.4). Appendix Table B.5 reports elasticities by manufacturing location. These elasticities describe procurement choices across products, rather than patients' price sensitivity for treatment.

[69] The estimated $FC_{\ell,\mathcal{T}}$ parameters thus measure location–period cost wedges relative to Foreign North and should not be interpreted as accounting measures of physical plant setup costs.

Supply estimates. The estimates quantify the trade-off between production costs and supply reliability. U.S. plants face marginal capacity costs nearly three times those in Foreign South and have a higher mean demand-to-capacity ratio. Yet Foreign South has a less favorable capacity-yield distribution, with a mean yield of 0.844 compared with 0.892 at Home. At the estimated prices and capacity choices, the mean share of unmet demand is 11.9% in Foreign South and 8.8% at Home (Appendix Table B.7). The Foreign South–Home fixed-cost index difference declines from 7.208 in 2005–2010 to -0.158 in 2017–2019, although the final-period difference is not statistically distinguishable from zero (Appendix Table B.6).

These estimates discipline the counterfactuals in Section 6, which compare reshoring subsidies, shortage-state price premia, and failure-to-supply penalties. All three policies induce new price and capacity choices; only the reshoring counterfactual allows production locations to change.[70]

6. Counterfactual Simulations

6.1. Objectives and Summary

Could we reduce shortages, and at what cost? I use the estimated model to evaluate three policies that have appeared in the debate over drug shortages. The first targets the global organization of production by subsidizing domestic manufacturing, thus promoting reshoring. The second allows prices to rise during shortages through a market-wide premium on delivered units, a stylized form of spot pricing. The third changes procurement incentives by imposing contractual failure-to-supply penalties on missing units.

These experiments compare bringing production home with adapting procurement contracts to the existing global supply chain. The reshoring subsidy changes location incentives, whereas shortage-contingent pricing and penalties hold observed production locations fixed and change firms' price and capacity choices. All three policies are anticipated before prices and capacity are chosen.

The counterfactuals point to one conclusion: effective procurement reform strengthens ex-ante incentives for resilient capacity, rather than simply paying more when supply is scarce. Shortage-contingent pricing likewise does little to reduce unmet demand, because the premium induces opposing capacity responses across suppliers. Failure-to-supply penalties instead raise the expected cost of each missing unit and induce larger buffers, substantially improving availability. These improvements come at higher contract prices: the access gains nearly offset the price effect in buyer surplus, with small positive gains when fines are rebated at the benchmark penalty rate. All simulations use the same estimated structural parameters and the same status-quo baseline, which reproduces the observed allocation over 2017 to 2019 at a policy rate of zero; outcomes are market-size weighted averages across market-years.

[70] In the reshoring counterfactual, location shares are expected choice probabilities from (38), not deterministic counts of baseline plant locations.

6.2. Counterfactual Changes in Realized Consumer Surplus

Consumer welfare in the counterfactuals is measured by the realized logit surplus index derived in Equation (23). For a given market (m, t), realized consumer surplus (CS) is

$$CS^{\text{real}}(\delta, s_0^{\text{real}}) = \underbrace{CS^{\text{logit}}(\delta)}_{\text{frictionless logit surplus}} - \underbrace{\left[s_0^{\text{real}} - s_0^D(\delta)\right] \bar{\mathcal{L}}(\delta)}_{\text{shortage penalty}}, \tag{40}$$

where $CS^{\text{logit}}(\delta)$ is the logit surplus index in (16), $s_0^D(\delta)$ is the latent outside share implied by demand, s_0^{real} is the realized outside share after supply shocks, and $\bar{\mathcal{L}}(\delta)$ is the average shortage loss in (22). The second term captures the welfare loss from buyers who would have purchased an inside good absent the supply shock but are displaced into the outside option. I call the difference between the realized and the latent outside share the forced outside share.

Let $(\delta_0, s_{0,0}^{\text{real}})$ denote the baseline allocation and $(\delta_1, s_{0,1}^{\text{real}})$ a counterfactual allocation, where the realized outside shares are the corresponding expected realized outside shares under each allocation. Following (40), the change in realized consumer surplus can be decomposed as

$$\begin{aligned}
\Delta CS^{\text{real}} &\equiv CS^{\text{real}}(\delta_1, s_{0,1}^{\text{real}}) - CS^{\text{real}}(\delta_0, s_{0,0}^{\text{real}}) \\
&= \underbrace{\left(\left[s_{0,0}^{\text{real}} - s_0^D(\delta_0)\right] - \left[s_{0,1}^{\text{real}} - s_0^D(\delta_1)\right] \right) \bar{\mathcal{L}}(\delta_0)}_{\Delta CS_{\text{access}}\ \text{(access effect)}} \\
&\quad + \underbrace{CS^{\text{logit}}(\delta_1) - CS^{\text{logit}}(\delta_0) - \left[s_{0,1}^{\text{real}} - s_0^D(\delta_1)\right] \left[\bar{\mathcal{L}}(\delta_1) - \bar{\mathcal{L}}(\delta_0)\right]}_{\Delta CS_{\text{price}}\ \text{(price/mean-utility effect)}}.
\end{aligned} \tag{41}$$

The access component values the reduction in shortage-induced displacement at the baseline loss per displaced buyer. The price component captures changes in logit surplus and in the loss associated with each displaced buyer. These components sum to the change in realized consumer surplus; improved access need not outweigh the effect of higher prices. Appendix C.1.1 gives the derivation.

Incidence of policy transfers. Each policy involves a transfer that affects what buyers pay without changing the allocation at which I value that transfer. I therefore report realized consumer surplus both at contract prices and at the price paid, charging buyers for subsidies and shortage-state premia or crediting them with fines rebated in full. The proportional-price valuation and aggregation are described in Appendix C.1.4. I also express each welfare change as the uniform change in baseline contract prices that buyers would value equally at their pre-policy purchases.[71]

6.3. Counterfactual 1: Reshoring Subsidies

In this counterfactual, I evaluate a policy that subsidizes reshoring of injectable drugs production to the U.S., in the spirit of the 2020 *Domestic Medical and Drug Manufacturing Tax Credit* proposal and the 2021

[71] Under the penalty, fines are returned in full to the buyers who contracted with the failing supplier. These measures compare outcomes from the buyers' side, not total welfare; Appendix C.1.4 reports transfers and manufacturers' profits separately.

American Jobs Plan.[72] The 2020 proposal would provide a tax credit of 10.5% of net income from eligible domestic production.[73] Policies focused on bringing back production of essential manufacturing sectors home have been widely discussed since 2017, both in the U.S. and Europe; see Appendix C.2 for institutional details.[74]

Reshoring Subsidies Framework. This counterfactual sets a subsidy payable to any firm that locates production in the U.S. (location $\ell = H$). Let s_{reshore} denote the subsidy rate. Under the subsidy regime, supplier j's location problem of equation (Eq. location) for product m in period t becomes

$$\ell^*_{jmt} \in \arg \max_{\ell \in \{H, F_N, F_S\}} \left\{ \mathbb{E}_{\omega \sim F_\ell} \left[\pi^\ell_{jmt} \right] - FC_{\ell, \mathcal{T}(t)} + \epsilon^\ell_{jmt} + s_{\text{reshore}} \, \mathbb{1}\{\ell = H\} \, \mathrm{R}^H_{jmt} \right\}, \qquad \text{(Eq. Reshoring)}$$

where $\mathbb{E}_{\omega \sim F_\ell}[\pi^\ell_{jmt}]$ is expected variable profit at the candidate-location price and capacity equilibrium, and $FC_{\ell,\mathcal{T}(t)}$ and ϵ^ℓ_{jmt} are the fixed-cost component and location-profitability shock of equation (Eq. FC), in millions of dollars, with the fixed-cost component varying across the periods defined in Section 5.3.3. The revenue base for the Home subsidy is expected realized revenue under candidate Home production:

$$\mathrm{R}^H_{jmt} \equiv \mathbb{E}_{\omega \sim F_H} \left[p^H_{jmt} q^{\text{real},H}_{jmt}(\omega) \right] = p^H_{jmt} \left(\left[1 - F_H(\theta^H_{jmt}) \right] Q^D_{jmt}(\delta^H) + \kappa^H_{jmt} h_H(\theta^H_{jmt}) \right), \tag{42}$$

with $q^{\text{real},H}_{jmt}(\omega) = \min\{Q^D_{jmt}(\delta^H), \kappa^H_{jmt}\omega\}$, $\theta^H_{jmt} = Q^D_{jmt}(\delta^H)/\kappa^H_{jmt}$ and $h_H(\theta) = \int_0^\theta \omega f_H(\omega)\, d\omega$, all evaluated at the Step 2 price–capacity equilibrium conditional on Home production. The subsidy acts as a location-specific shift in the Home profit index, making the U.S. relatively more attractive as a production location; it does not enter the pricing and capacity conditions of Section 4.4.2, so a plant that moves home chooses its buffer exactly as any Home plant does. A policy with $s_{\text{reshore}} = 0.105$ corresponds to a revenue-equivalent 10.5% subsidy for domestic production. For values of $s_{\text{reshore}} \in [0, 0.20]$, I re-solve the location problem for every plant and then the pricing–capacity game in every market given the new distribution of plants across regions, holding all estimated structural parameters fixed.

Results. Figure 6 summarizes how equilibrium outcomes change as I vary the reshoring subsidy between 0 (baseline) and 20% of expected Home revenue (the benchmark subsidy rate is 10.5%); Appendix Figure C.1 reports additional outcomes. The y-axes report percentage deviations from the baseline.

First, the subsidy induces a substantial reallocation of production to the U.S., but fails to reduce shortages. A 10.5% subsidy doubles the expected share of product lines made in the United States, from 12 to 25%, and a 20% subsidy brings it to one half (Appendix panel C.1d). Yet panel 6b shows that the probability of a shortage is unchanged at the benchmark rate and slightly higher at a 20% subsidy, while the expected share of excess demand rises, by 5% at the benchmark rate and by 40% at the largest, and the share of patients forced into the outside option rises by 14% at the benchmark rate

[72] See the 2020 U.S. House Ways and Means Committee, "U.S. can boost domestic production of essential medicines and their ingredients with tax incentives," and the 2021 *American Jobs Plan* of Joe Biden: "Building Resilient Supply Chains, Revitalizing American Manufacturing and Fostering Broad-Based Growth".

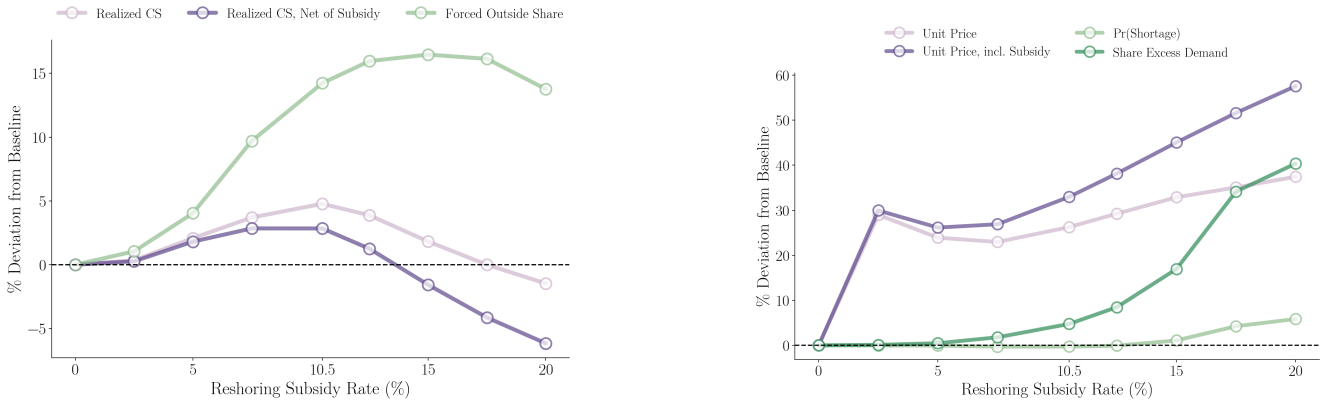
[73] The counterfactual uses a subsidy indexed to expected Home revenue.

[74] See, for instance, Washington Post, January 2023 or The Economist Report, 2021.

(panel 6a). Moving production home changes both production costs and disruption exposure. Once manufacturers adjust prices and capacity to these new conditions, the reallocation does not reduce the expected share of unmet demand.

Second, reshoring is *costly* in terms of prices. Panel 6b shows that the products that relocate to the U.S. charge about a quarter more than they did abroad at the benchmark rate, and a third more once buyers finance the subsidy through the prices they pay; at a 20% subsidy the increase reaches 58%. Even averaged over all products, most of which do not move, prices rise by 8% at the benchmark rate and by 31% at a 20% subsidy once the financing of the subsidy is taken into account (Appendix panel C.1a).

Third, reshoring yields only modest gains in consumer welfare. Panel 6a shows that realized consumer surplus increases by 4.8% at the benchmark rate. The gain does not come from better access, since more patients are forced into the outside option: the decomposition in Appendix panel C.1c attributes all of it to the price effect of (41). It comes from a few large products that move from other high-income countries to the U.S., where production costs are lower, and whose prices fall by more than half, while the many products that move from Asia nearly double theirs (Appendix Figure C.2).[75] When buyers finance the subsidy, worth 1.7 billion dollars a year at the benchmark rate, the gain falls to 2.8%. In price terms, buyers value the gain at the benchmark rate about as much as a 17% cut in all contract prices at their pre-policy purchases, and 11% once they finance the subsidy. At a 20% subsidy, which costs 12 billion dollars a year, both lines turn negative, at −1.5 and −6.2%.



(a) *Realized CS and Forced Outside Share* (b) *Unit Prices of Reshored Products and Shortages*

Figure 6: **Changes in Equilibrium Outcomes as a Function of Reshoring Subsidy Levels**

Notes: The y-axis displays changes in equilibrium outcomes, measured in percentage deviation from the baseline; the x-axis reports the reshoring subsidy rate in percent of Home revenue (the baseline is 0; the benchmark subsidy rate is 10.5%). Panel 6a: realized consumer surplus at contract prices, realized consumer surplus at the price paid with the subsidy's outlay charged to buyers, and the share of potential buyers forced into the outside option. Panel 6b: average unit price of the products that relocate to the U.S., with and without the subsidy's financing charged to buyers as a proportional surcharge in the same market (purple), the supplier-level probability of a shortage and the market-level expected share of excess demand (green). 2017 to 2019, weighted by market size. Additional outcomes are reported in Figure C.1.

[75]The gain is concentrated. Five market-years, enoxaparin in each year and ceftriaxone in 2018 and 2019, hold 3% of potential demand but nearly half of the welfare weight, because their latent outside shares are small and the surplus loss per rationed buyer correspondingly large. Without them realized consumer surplus at the benchmark rate rises by 0.6% and realized consumer surplus at the price paid is within half a percent of the baseline; at a 20% subsidy both losses deepen, to 5 and 7%. The shortage results do not depend on these markets.

Conclusion. Overall, subsidizing reshoring is a *costly* way to address shortages in this market. It generates a sizable reallocation of manufacturing towards the U.S. and large price increases for the products that move, but no improvement in shortage outcomes: the probability of shortages does not decline and the expected share of excess demand rises for every positive subsidy rate. The subsidy changes where drugs are made, not what spare capacity earns. U.S. plants produce at higher cost than Asian plants and keep less capacity above the demand they contract, and a drug that moves home faces both. The counterfactual does not contradict the reduced-form evidence that offshoring raises shortages. It shows that moving production home trades larger supply shocks for thinner buffers: what determines reliability is the return to holding spare capacity, not the location of the plant.[76]

6.4. Counterfactual 2: Shortage-Contingent Pricing

The second counterfactual targets price rigidities in procurement contracts by allowing short-run transaction prices to respond to realized supply conditions. In the baseline environment, unit prices p_{jmt} are fixed over the one-year contract horizon: manufacturers set contract prices and capacity at the beginning of period t, and these terms are honored regardless of subsequent supply shortfalls. Here, I consider a simple shortage-contingent pricing rule: a market-wide price premium on delivered units in shortage states.[77] Because manufacturers anticipate the rule when they set contract prices and capacity, it changes their ex-ante incentives as well as the price paid after a shock.[78]

Shortage-contingent pricing framework. Under this rule, each supplier j still chooses a contract price p_{jmt} and a capacity κ_{jmt} at the beginning of period t, before yield shocks realize, and buyers allocate demand at contract prices. Let $I^{\text{short}}_{mt}(\boldsymbol{\omega}) = \mathbb{1}\{\exists\, k \in J_{mt} : \kappa_{kmt}\omega_{kmt} < Q^{D}_{kmt}(\boldsymbol{\delta})\}$ indicate that at least one supplier in market (m,t) is short after yield shocks realize, in the sense of equation (7). The realized transaction price is then

$$\tilde{p}_{jmt}(\boldsymbol{\omega}) = p_{jmt}\Big(1 + \psi^{\text{spot}}\, I^{\text{short}}_{mt}(\boldsymbol{\omega})\Big), \qquad \text{(Eq. Spot Price)}$$

where $\psi^{\text{spot}} \geq 0$ is the premium paid in shortage states: a value of 0.25 pays delivered units 125% of the contract price when the market is short, and a value of one doubles their price. The realized transaction price is thus a random variable that depends on the realization of shortages in the market. The baseline corresponds to $\psi^{\text{spot}} = 0$; in the simulations I vary ψ^{spot} between 0 and 1.25. The rule changes only the

[76] It is also important to acknowledge that the consumer welfare measures reported here abstract from the real-world frictions of relocation. Transferring a single drug product to a new site can take up to two years, and building a new facility can require investments on the order of $2 billion and a timeline of five years. As they ignore these adjustment costs, the counterfactual results are conservative.

[77] Yurukoglu, Liebman, and Ridley (2017) note that generic sterile injectables share with electricity generation the features that make shortages costly: timing is critical, storage is costly, and products are undifferentiated, and that electricity markets address shortages with a mixture of rapid price adjustment and regulation of capacity. The analogy stops at the short run. A price spike in electricity calls idle generators online within minutes; in sterile injectables, a line that fails a quality inspection cannot produce at any price and other lines run near capacity, so a shortage-contingent premium rewards the units delivered but cannot relax the capacity constraint, which is what the simulations show.

[78] The implicit assumption is that, in states without excess demand, unused capacity has no economic payoff; any surplus capacity is disposed of at the end of period t.

price received in each realized state; it does not relax the within-period capacity constraint, and firms can sell at most their realized available supply in any state.

Taking expectations over yield shocks, a supplier's expected profit is the baseline expected profit of Section 4.4.2 plus the expected premium on the units it delivers in shortage states,

$$\mathbb{E}_{\omega\sim F_\ell}\left[\pi_{jmt}^{\ell,\text{spot}}\right] = \mathbb{E}_{\omega\sim F_\ell}\left[\pi_{jmt}^{\ell}\right] + \psi^{\text{spot}}\, p_{jmt}\, \mathbb{E}\left[q_{jmt}^{\text{real}}\, I_{mt}^{\text{short}}\right]. \qquad \text{(Step-2 Spot)}$$

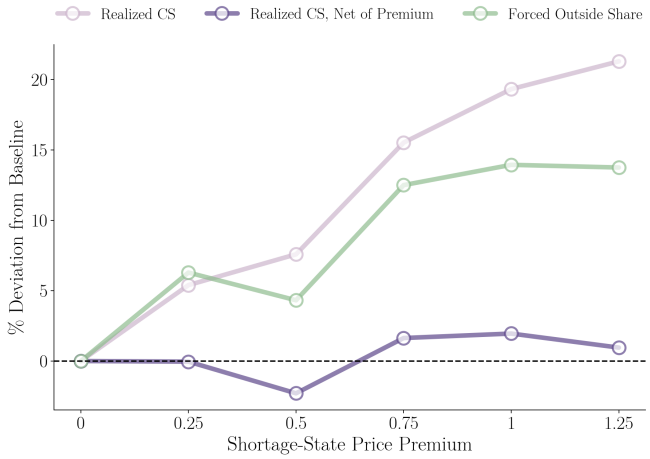
The expectation in the premium term is joint, because the shortage indicator depends on rivals' yield shocks as well as on firm j's own; Appendix C.3 derives it in closed form under independent plant-level shocks. The premium pulls the capacity choice in two directions. A larger buffer delivers more units at the premium when a rival is short, but it also removes the states in which supplier j is itself short and the premium is paid at all. For each ψ^{spot}, I compute the Nash equilibrium of the pricing–capacity game in every market by the same fixed-point procedure as in the baseline model, holding production locations at their observed values.

Results. Figure 7 summarizes how equilibrium outcomes change as I vary ψ^{spot} between 0 (baseline) and 1.25; Appendix Figure C.3 reports additional outcomes.

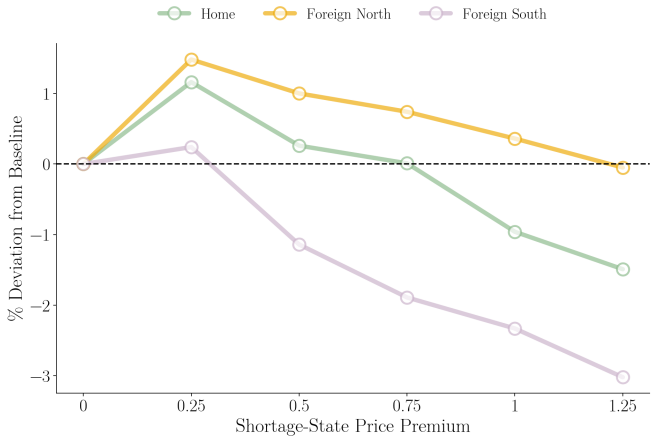
Three main patterns emerge. First, shortage-contingent pricing has little effect on the expected share of unmet demand but raises shortage incidence. At a premium of 0.25, the supplier shortage probability rises by 31.8%, while the expected share of excess demand rises by 0.5% (Table 2). Most suppliers build larger buffers, but about one in six gives up its buffer entirely. The increase in shortage incidence therefore reflects capacity choices made before shocks, rather than withholding available output afterward.

Second, the gains in consumer surplus at contract prices do not come from improved access and are largely offset once buyers pay the premium. At premiums of 0.25 and 0.50, realized consumer surplus at contract prices rises by about 5 to 8 percent as manufacturers lower their bids in anticipation of premium revenue. At the price paid, the corresponding changes are approximately zero and -2.3%. At a premium of one, surplus at the price paid is 2.0% above baseline, but the forced outside share is 13.9% higher. Allowing prices to rise during shortages thus need not improve availability, even when it raises buyer surplus.

Third, the incentive to hold a buffer differs across production regions. In relatively safe locations, a supplier without a buffer earns the premium whenever its yield falls below one, while losing relatively few sales. A quarter to a third of suppliers at Home and in Foreign North give up their buffers, compared with about a tenth in Foreign South (Appendix Figure C.3). Shortage probability rises by roughly one half at Home and three quarters in Foreign North, compared with about 15 percent in Foreign South (Appendix Figure C.3). Suppliers that retain a buffer generally expand it, so the rule changes the distribution of shortage risk rather than uniformly improving reliability.



(a) *Realized CS and Forced Outside Share*



(b) *Share of Excess Demand by Production Region*

Figure 7: **Changes in Equilibrium Outcomes as a Function of the Shortage-State Premium**

Notes: The y-axis displays changes in equilibrium outcomes, measured in percentage deviation from the baseline; the x-axis displays the shortage-state premium ψ^{spot} from equation (Eq. Spot Price) (the baseline is 0); production locations are held at their observed values. Panel 7a: realized consumer surplus at contract prices, realized consumer surplus at the price paid with the premium paid by buyers, and the share of potential buyers forced into the outside option, weighted by market size. Panel 7b: expected share of a supplier's demand that goes unmet, averaged across suppliers within each production region; Home is the U.S., Foreign North other high-income locations, Foreign South Asia. 2017 to 2019. Additional outcomes are reported in Appendix Figure C.3.

Conclusion. Allowing prices to rise when supply is scarce does not by itself resolve the shortage problem. The premium is anticipated when manufacturers choose capacity, but it rewards deliveries only while at least one supplier remains short. As a result, it encourages some suppliers to expand their buffers and others to give them up, leaving unmet demand largely unchanged and shortages more frequent.[79]

6.5. Counterfactual 3: Failure-to-Supply Penalties

The third counterfactual targets the lack of financial incentives for manufacturers to invest in resilient capacity *ex ante*. In the baseline procurement environment, contracts impose little cost on manufacturers when realized supply falls short of latent demand.[80] As a result, the private expected cost of a supply shortfall is below its welfare cost to buyers: resilient capacity may be socially valuable but

[79]The experiment maintains contracted allocations after shocks. It does not evaluate a market-clearing spot market in which buyers can switch suppliers.

[80]Group purchasing contracts sometimes contain failure-to-supply clauses, but they are narrow (Haninger, Jessup, and Koehler, 2011), and Todd Ebert, then president of the Healthcare Supply Chain Association, told me in an interview on February 22, 2022 that they are almost never invoked in practice. Section 2-615 of the Uniform Commercial Code holds that failing to deliver a contracted product is not a breach when the product is unavailable in the market (§2-615 excuses a seller when an unforeseen event makes delivery commercially impracticable), and the duration of compensation under these clauses has narrowed over time. The erosion of these clauses is consistent with the evidence that shortages of injectable products were rare and stable before the current crisis. Suppliers continue selling products that went short at nearly the baseline rate, and I find no evidence of a subsequent decline in their national market shares once supply has recovered (Appendix D.4). Appendix D.1 details contracting practices.

privately under-priced.[81] I therefore evaluate a policy that forces manufacturers to internalize the cost of disruptions: a simple failure-to-supply penalty that makes unreliability privately costly. This design is in the spirit of "Japanese-type" procurement systems (Taylor and Wiggins, 1997; Heise et al., 2024), in which buyers commit to long-term relationships at a premium, and suppliers that fail to deliver lose the relationship.[82] This mechanism acts as a pre-commitment device: the contractual penalty raises the expected cost of a supply shortfall, strengthening manufacturers' incentives to invest in capacity ex ante.

Penalty framework. A manufacturer j in market m and period t now incurs a penalty whenever a shortage occurs; that is, whenever realized supply is insufficient to meet contracted demand. The penalty term $\mathcal{P}_{jmt}$ is proportional to the revenue supplier j would have earned on the missing units:

$$\mathcal{P}_{jmt} \equiv \psi^{\text{pen}}\, p_{jmt}\, Q^{\text{short}}_{jmt}, \qquad Q^{\text{short}}_{jmt} \equiv \max\{Q^{D}_{jmt}(\delta) - \kappa_{jmt}\,\omega_{jmt}, 0\}, \tag{43}$$

where $\psi^{\text{pen}} \geq 0$ is a dimensionless penalty rate. When $\psi^{\text{pen}} = 1$, a firm that completely fails to deliver its contracted quantity pays a fine equal to the gross revenue it would have earned on those units; for partial shortfalls, the penalty scales linearly with the shortage volume. Using the baseline expected profit expression from Section 4.4.2, expected profit under the penalty scheme is

$$\begin{aligned}\mathbb{E}_{\omega\sim F_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\right] = \underbrace{p_{jmt}\Big(\left[1 - F_\ell(\theta_{jmt})\right] Q^{D}_{jmt}(\delta) + \kappa_{jmt}\, h_\ell(\theta_{jmt})\Big) - c^{\ell}_{jmt}\,\kappa_{jmt}}_{\text{Baseline expected profit } \mathbb{E}_{\omega\sim F_\ell}[\pi^{\ell}_{jmt}]} \\ \underbrace{- \psi^{\text{pen}}\, p_{jmt}\Big(Q^{D}_{jmt}(\delta)\, F_\ell(\theta_{jmt}) - \kappa_{jmt}\, h_\ell(\theta_{jmt})\Big)}_{\text{Expected penalty } \mathbb{E}_{\omega\sim F_\ell}[\mathcal{P}_{jmt}]},\end{aligned} \tag{44}$$

The expected penalty is $\psi^{\text{pen}} p_{jmt}$ times the expected quantity of unmet demand,

$$\mathbb{E}_{\omega\sim F_\ell}\left[Q^{\text{short}}_{jmt}\right] = Q^{D}_{jmt}(\delta) F_\ell(\theta_{jmt}) - \kappa_{jmt} h_\ell(\theta_{jmt}), \tag{45}$$

The demand-to-capacity ratio θ_{jmt} and truncated yield moment h_ℓ are defined in equations (6) and (13). At $\psi^{\text{pen}} = 0$, equation (44) reduces to baseline expected profit. A positive penalty increases the marginal benefit of capacity by the expected fine avoided on additional deliveries (Appendix C.4). For each $\psi^{\text{pen}} \in [0, 2]$, manufacturers choose prices and capacities simultaneously at their observed production locations. For example, if expected unmet demand is 20% of contracted demand, a penalty rate of 0.5 implies an expected fine equal to 10% of the value of contracted demand.

[81] This wedge may reflect two frictions. First, resilience investments (such as maintaining buffer capacity) are effectively public goods in a common-agency setting: if one buyer pays a premium to support a backup line, it cannot exclude other buyers from benefiting if a shortage is averted, leading to a free-rider problem. Second, information asymmetries regarding plant-level manufacturing quality prevent buyers from effectively screening for resilient suppliers at the bidding stage.

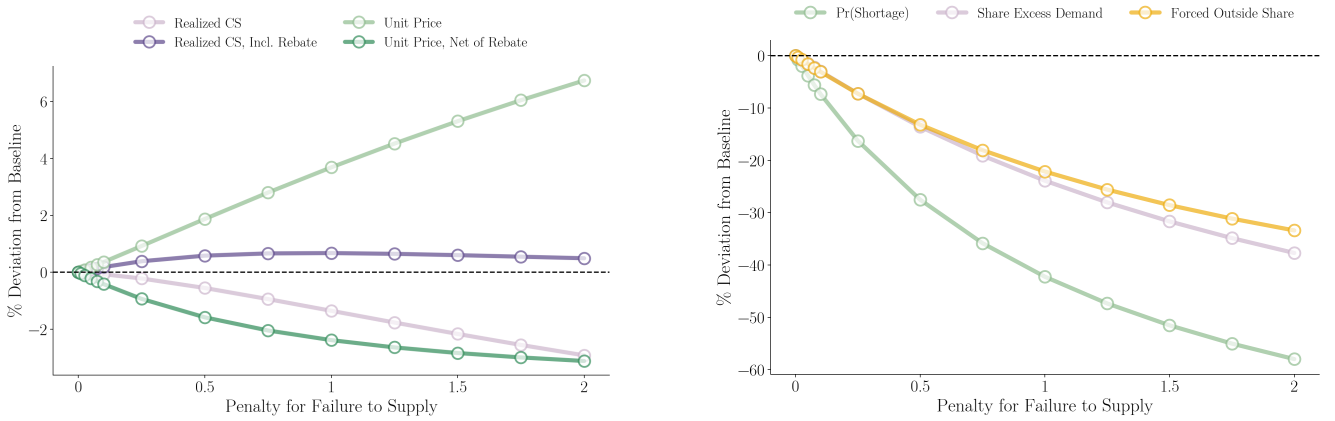
[82] Taylor and Wiggins (1997) and Heise et al. (2024) contrast these "Japanese" contracts with the prevailing "American" system, where buyers rely on frequent competitive bidding and costly inspection to discipline quality. In the generic sterile injectable market, inspections are especially difficult for foreign plants and buyers cannot easily observe plant-level reliability, so reliability investments are largely a hidden action. In addition, hospitals contract through several GPOs that negotiate separately, which creates a common-agency free-rider problem: any buyer that negotiates strict penalties confers benefits to all other buyers when a disruption occurs. A regulator or consortium-backed failure-to-supply clause can be viewed as a coordination device that raises the expected cost of shortfalls in all contracts simultaneously.

Results. Figure 8 presents the simulation results, summarizing how equilibrium outcomes change with the penalty rate. The y-axis reports percentage deviations in equilibrium outcomes from baseline, and the x-axis plots the penalty rate ψ^{pen}. Appendix Figure C.5 reports the capacity response and the welfare decomposition.

First, failure-to-supply penalties substantially reduce shortages. A rate of 0.5 lowers the probability of a supplier shortage by 27.5%, the expected share of excess demand by 13.6%, and the forced outside share by 13.2%. At a rate of one, the reductions are 42.2%, 23.9%, and 22.1% (Table 2). Manufacturers respond to the higher cost of missing deliveries by expanding capacity before shocks occur: planned capacity rises by 13% and 25% at the two rates (Appendix Figure C.5).

Second, the improvement in availability comes at higher contract prices. The demand-weighted contract price rises by 9% at a rate of 0.5 and 18% at a rate of one. The corresponding increases in average contract price per delivered unit are 1.9% and 3.7%, a different statistic that uses the quantities delivered after the policy. Once fines are rebated, the latter price measure falls by 1.6% and 2.4%, respectively.

Third, the access gains nearly offset the price effect in consumer surplus. At a rate of 0.5, improved access adds 1.6% of baseline realized surplus, while the price effect subtracts 2.2%, leaving surplus at contract prices 0.6% below baseline. Returning the fines to buyers makes the price-paid surplus index 0.6% higher than baseline. At a rate of one, the corresponding changes are -1.4% and $+0.7\%$. Thus, the penalty delivers a substantial improvement in availability with a small change in buyer surplus, rather than a large welfare gain.



(a) *Realized CS and Unit Prices* (b) *Shortage Outcomes*

Figure 8: **Changes in Equilibrium Outcomes under Failure-to-Supply Penalties**

Notes: The y-axis displays changes in equilibrium outcomes, measured in percentage deviation from the baseline; the x-axis reports the penalty rate ψ^{pen} from equation (43) ($\psi^{\text{pen}} = 0$ is the baseline); production locations are held at their observed values. Panel 8a: realized consumer surplus at contract prices and realized consumer surplus at the price paid with the fines rebated to buyers (purple), and the average unit price paid per delivered unit with and without the rebate (green); the darker line of each pair includes the rebate. Panel 8b: supplier-level probability of a shortage, market-level expected share of excess demand, and share of potential buyers forced into the outside option. 2017 to 2019, weighted by market size. Additional outcomes are reported in Appendix Figure C.5.

Conclusion. Overall, this counterfactual illustrates that a contract design which raises the private cost of failure to supply delivers sizeable improvements in product availability at a moderate price.

Table 2: **Counterfactual Changes in Shortages, Prices and Consumer Surplus**

	Reshoring			Shortage-contingent pricing			Failure-to-supply penalty		
	5%	10.5%	20%	0.25	0.50	1	0.25	0.50	1
Shortages, percent change									
Supplier shortage probability	−0.1	−0.3	+5.9	+31.8	+30.8	+29.9	−16.3	−27.5	−42.2
Expected share of excess demand	+0.5	+4.8	+40.4	+0.5	−0.8	−1.9	−7.2	−13.6	−23.9
Forced outside share	+4.0	+14.2	+13.8	+6.3	+4.3	+13.9	−7.3	−13.2	−22.1
Prices, percent change									
Per-unit price, excl. transfer§	+23.9	+26.3	+37.4	−1.7	−3.3	−8.1	+0.9	+1.9	+3.7
Per-unit price incl. transfer§	+26.1	+33.0	+57.6	+22.8	+45.0	+83.7	−0.9	−1.6	−2.4
Realized consumer surplus, percent of baseline									
At contract prices	+2.1	+4.8	−1.5	+5.4	+7.6	+19.3	−0.2	−0.6	−1.4
At the price paid, net of transfer	+1.8	+2.8	−6.2	−0.0	−2.3	+2.0	+0.4	+0.6	+0.7
Transfer per year, billion dollars	0.5	1.7	11.7	12.7	26.2	54.9	1.5	2.8	4.9

Notes: Percentage deviations from the baseline over 2017 to 2019, weighted by market size; columns give the reshoring subsidy rate in percent of Home revenue, the shortage-state premium ψ^{spot} and the penalty rate ψ^{pen}. §For reshoring, the average contract price of the products that relocate to the U.S.; for shortage-contingent pricing and the penalty, the average price buyers pay per delivered unit, weighting products by units delivered after the policy. The transfer is the subsidy's outlay (charged to buyers in the second welfare line), the premium paid by buyers, or the fines rebated to buyers.

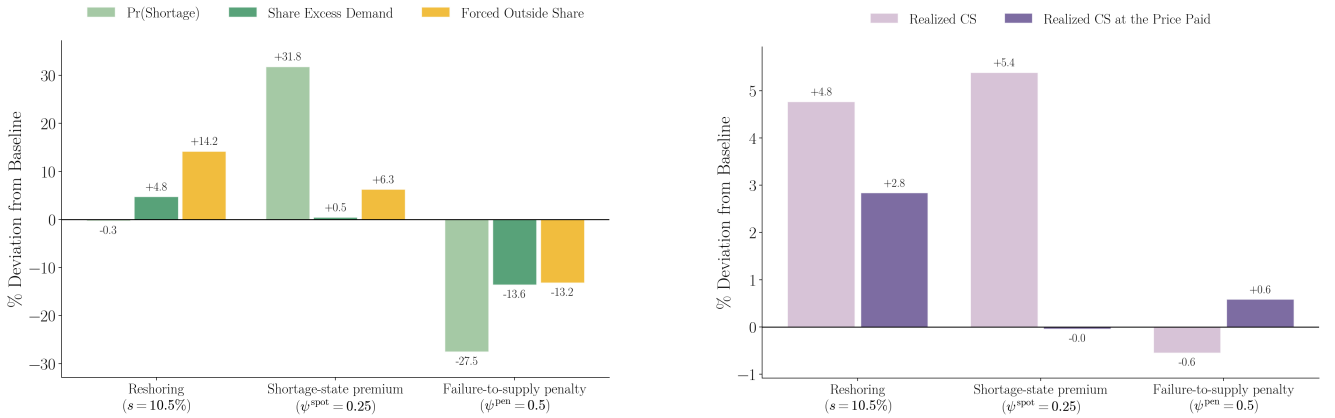
Shortage-contingent pricing rewards deliveries when the market is short, but creates mixed incentives for capacity investment: most suppliers build larger buffers, while some give them up. Failure-to-supply penalties instead increase manufacturers' losses on each missing unit, inducing larger buffers before disruptions occur and reducing the fraction of patients rationed into the outside option.[83] Consumer welfare at contract prices is roughly unchanged, and buyers gain once the fines are returned to them. There is inevitably a trade-off between the price buyers are willing to pay for a drug and how reliable they want its supply to be.

6.6. Counterfactuals: Comparison and Takeaways

This section compares the three policies in terms of shortages, prices, and realized consumer surplus. Table 2 reports selected rates, and Figure 9 summarizes the benchmark comparisons. The rates apply to different payments, so equal numerical rates do not represent equal policy costs.

Consumer welfare and shortages. At the benchmark rates, only the failure-to-supply penalty substantially reduces both shortage incidence and unmet demand. A penalty at a rate of one half lowers the probability of a shortage by 28% and the share of excess demand by 14%; the premium raises the probability of a shortage by about a third at any rate, and the subsidy leaves it unchanged while

[83]Civica Rx provides a related example of longer-term purchasing commitments intended to secure reliable supply. Its five-year agreement with Hikma combined guaranteed demand with pricing intended to support production planning and investment in manufacturing, illustrating how buyers' commitments can support suppliers' investments in reliability; see Civica Rx (2019).



(a) *Shortage Outcomes* (b) *Realized Consumer Surplus*

Figure 9: **Comparing the Three Policies at Benchmark Rates**

Notes: Percentage deviations from the baseline at a reshoring subsidy of 10.5%, a shortage-state premium of 0.25 and a failure-to-supply penalty rate of 0.5, the columns of Table 2. Panel 9a: supplier-level probability of a shortage, expected share of excess demand, and forced outside share. Panel 9b: realized consumer surplus at contract prices and at the price paid.

raising the share of excess demand (panel 9a). The decomposition of equation (41) shows that the penalty is also the only policy with a positive access component at these rates. Under reshoring and shortage-contingent pricing, realized consumer surplus at contract prices rises, by 4.8% at the benchmark subsidy and by 5% at a premium of one quarter, but the access component is negative in both cases: more patients are forced into the outside option, and the gains come entirely from lower contract prices, concentrated in a few large markets under reshoring and taken back by the premium under shortage-contingent pricing. Under the penalty, the access component is positive, as fewer patients are rationed into the outside option, while the price component is negative, as suppliers charge for the buffers they now hold; at a rate of one half the two nearly cancel at contract prices, and once the fines are returned to buyers the penalty is the only instrument that delivers both fewer shortages and a gain in consumer surplus, of under one percent (panel 9b). All three policies leave some demand unmet at the rates considered.

What it costs to reduce shortages. How large a penalty does it take to reduce shortages by a given amount, and what does it cost buyers? Appendix Table C.2 answers for the two shortage measures and at a common annual transfer amount. Cutting the expected share of excess demand by a tenth requires a penalty rate of 0.36, which raises the demand-weighted contract price by 6% and collects 2 billion dollars a year in fines that return to buyers; cutting it by a quarter requires a rate above one, a contract price a fifth higher and 5 billion dollars a year, and halving it is out of reach at any rate up to two. A quarter cut in the probability of a shortage requires a rate of 0.44 and 2.5 billion dollars a year, three percent of buyer spending, at which realized consumer surplus falls by half a percent at contract prices and rises by half a percent once the fines are returned. Neither of the other two instruments reaches any target, since the subsidy raises the share of excess demand and the premium the probability of a shortage at every rate. At a common annual transfer amount, 2.5 billion dollars a year in fines cuts the share of excess demand by 12% and the probability of a shortage by a quarter, whereas the same

amount paid out as a reshoring subsidy, a rate of 12%, raises the share of excess demand by 7.5% and leaves the probability of a shortage nearly unchanged; a premium of 0.25 already costs five times that amount.[84]

Prices and transfers. The three policies also differ in who pays (Appendix Table C.1). Reshoring raises the prices of the products that move by a quarter to a third, and the subsidy is an outlay of 2 to 12 billion dollars a year that accrues to manufacturers. Shortage-contingent pricing is a transfer from buyers to manufacturers of 13 billion dollars a year at a premium of one quarter, paid in shortage states: the average price buyers pay per delivered unit, premium included, rises by 23% while contract prices fall by 2%. Under failure-to-supply penalties, the demand-weighted contract price rises by 9 to 18% and the average contract price per delivered unit by 2 to 4%, while the price net of the rebate falls, and the fines, 3 to 5 billion dollars a year, are paid by the suppliers that fall short and returned to buyers.

Policy implications. The simulations suggest that improving procurement incentives can reduce shortages without reversing the global organization of production. Moving production home changes disruption exposure but also the cost of holding capacity; paying more when the market is short rewards deliveries, but encourages some suppliers to expand their buffers and others to give them up. Failure-to-supply penalties act more directly on the cost of missing units and deliver larger improvements in availability in these experiments. These gains are not costless: manufacturers charge higher contract prices, and the modest gains in buyer surplus depend on returning fines to buyers.[85] Effective procurement reform must therefore provide a return to reliable delivery while accounting for the price buyers pay for it. These results do not rule out a role for targeted reshoring of a narrow set of critical drugs alongside broader procurement reform.

7. Conclusion

What drives persistent shortages of generic injectable drugs? They reflect the interaction of production disruptions with a procurement system that has failed to adapt to the offshore structure of generic drug supply chains. Price and capacity rigidities have long been features of these markets. Offshoring changes the cost and risk of production, while limited transparency makes differences in reliability difficult to reward. By mapping injectable drugs to their manufacturing plants, I provide evidence that offshoring increases the incidence and duration of shortages.

I develop and estimate a structural model of global drug supply and procurement that quantifies manufacturers' trade-off between cost-cutting and supply reliability. The estimates show that Asian

[84] These amounts are transfers, not common measures of fiscal or resource cost: penalties collect funds from suppliers, whereas subsidies and shortage premia pay funds to suppliers.

[85] Policies that prevent the prices of these products from rising act on this margin. The Inflation Reduction Act of 2022 includes provisions to lower prescription drug costs for people with Medicare and to reduce federal drug spending, adopted amidst strong bipartisan public support for addressing high drug prices and in the context of collusive behavior in the non-injectable generic industry (Cuddy, 2020). Scott Gottlieb, former FDA commissioner, argued in a May 2023 CBS interview that "the administration under the IRA should carve out these old sterile injectable drugs entirely."

production offers lower costs but greater exposure to capacity losses. Production in the United States reduces that exposure, but makes spare capacity more expensive. To estimate the model, I develop a method to recover demand when shortages constrain sales. I also derive a closed-form expression for realized consumer surplus, accounting for the losses of buyers forced into an outside option they had rejected. The resulting welfare decomposition separates the benefits of improved access from the effects of prices and drug characteristics.

The counterfactuals compare changing production geography with changing procurement contracts at existing locations. Reshoring subsidies raise the average price of products that relocate without materially reducing shortages. When consumer surplus increases, the gains come from price reductions in a few large markets, not from improved access. Paying a premium on deliveries during shortages likewise raises what buyers pay without substantially reducing unmet demand. Failure-to-supply penalties instead induce larger capacity buffers by increasing the private cost of each missing unit. A penalty equal to half the contract price per missing unit reduces the expected share of unmet demand by about 14 percent while demand-weighted contract prices rise by 9 percent. The gains from improved access nearly offset the losses from higher contract prices, and consumer surplus rises modestly once fines are returned to buyers.

There is inevitably a trade-off between the price people are willing to pay for a drug and how reliable they want its supply to be. Investing in resilient capacity comes at a cost, but low prices are of little benefit to buyers who cannot obtain the drugs they chose. The challenge is to design procurement contracts that reward reliable capacity before disruptions occur, while retaining the production advantages of global supply chains.

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